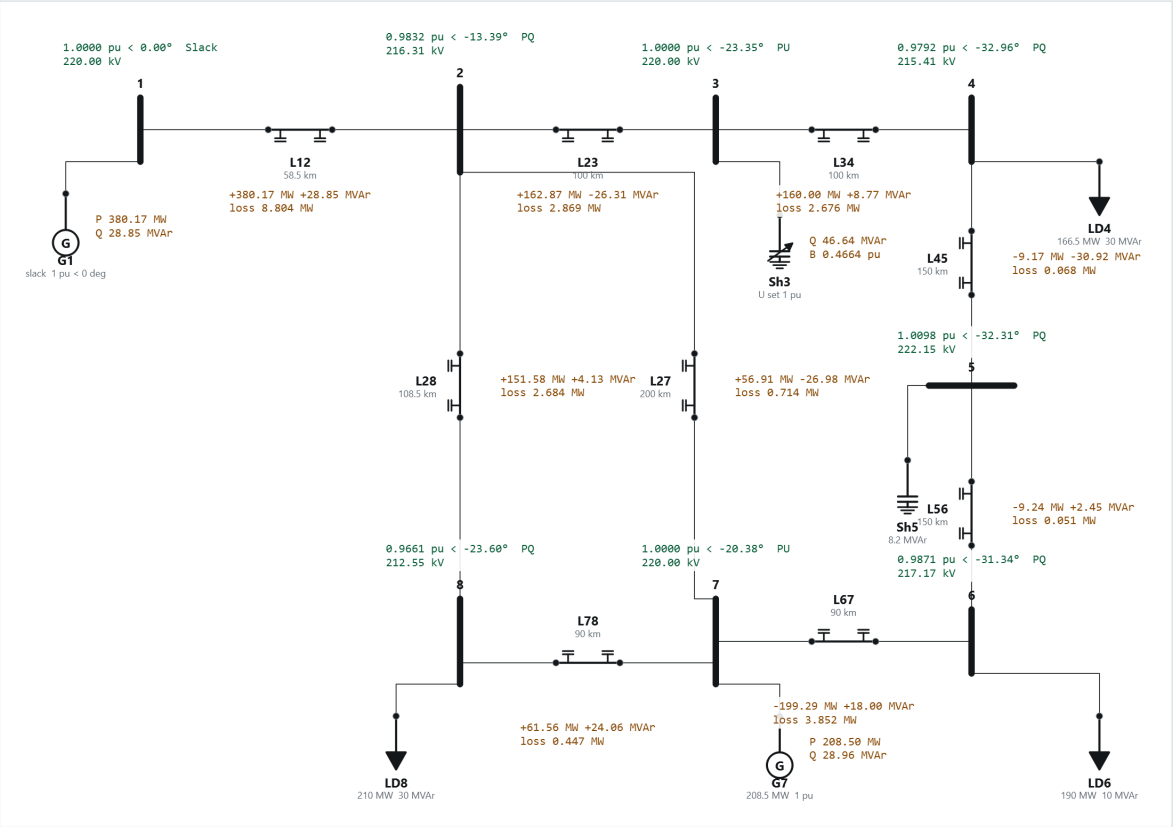
PowerLab

Interactive single-line diagrams and power system analysis

User Manual · Version 1.0.0



The Newton–Raphson load flow of the EG2100 Assignment LF network, as PowerLab annotates and exports it.

PowerLab grew out of the course EG2100 Power System Analysis (KTH). It draws a network on a canvas and solves, in per unit and with every number traceable: two-port matrices, equivalent impedances, bus voltages, the admittance matrices of the three sequence networks, Thevenin equivalents, short circuits of every type, unbalanced loads and the Newton–Raphson load flow.

github.com/akamaxe/powerlab

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Chapter 1

Introduction

PowerLab is an interactive single-line diagram editor with a power system analysis engine behind it. You draw a network on a canvas, much as in DIgSILENT PowerFactory, PSS/E or ETAP. PowerLab converts it to per unit and runs the studies a power system analysis course asks for, with the results printed the way the course prints them. It started from the KTH course *EG2100 Power System Analysis*. Version 1.0 solves all four course assignments (S1, S2, S3 and LF) and reproduces their reference solutions digit by digit.

1.1 What PowerLab does

Draw buses, external grids, generators, two-winding transformers, PI-model lines and cables, loads, shunt capacitors and reactors, static var compensators and raw impedances. Wires route themselves at right angles. Busbars resize, labels stay upright, and every edit can be undone (Chapters 4 and 5).

Parametrise any numeric field with an expression over project variables, such as `(217 + A)/1000`. Change `A` and the whole network is re-evaluated (Chapter 7).

Organise the work in *study cases*. A case is a study, its options and, optionally, a variant of the network: elements taken out of service and fields overridden by an expression. The drawn network is never changed (Chapter 8).

Analyse with eleven studies (Chapter 9):

- the per-unit model and the sequence model;
- two-port (ABCD) matrices, equivalent impedances and linear bus voltages;
- the positive-, negative- and zero-sequence admittance matrices;
- Thevenin equivalents in all three sequences;
- three-phase, single line-to-ground, line-to-line and double line-to-ground faults through fault and ground impedances;
- unbalanced loads;
- the Newton–Raphson load flow, with its iteration log and Jacobian.

Read each result in its own tab: a headline, warnings and tables formatted like MATLAB's `format short`. Annotate the diagram with any result and colour it by voltage level or bus voltage (Chapter 10).

Edit in bulk every element of a type in the data manager, a spreadsheet-like table (Chapter 11).

Export results as text, CSV, LaTeX tables or JSON, the diagram as PNG, SVG or PDF, and matrices as MATLAB literals that keep every digit (Chapter 12).

Automate from a terminal with `run_headless.py`, which runs a project's cases, sets variables and exports what it ran (Chapter 13).

1.2 Principles

Four ideas run through the program, and they explain most of its behaviour.

Glass box. Every number can be traced. The per-unit table shows each component's conversion with its base factor. The admittance, impedance and Jacobian matrices are printed with row and column labels, and the load flow logs every iteration.

Verified. Each study checks itself by a second, independent method. The Thevenin impedance is computed from the impedance matrix and again by Kron reduction. The two-port matrix comes from the element cascade and from the Y-bus. The load flow closes the power balance. The cross-check residuals are part of the results.

Parametric. Variables, expressions and study cases let one project hold every variant of a problem, and re-evaluate it when a parameter changes.

Course-grade presentation. Impedance and complex power are printed in rectangular form, voltage and current in polar form with the angle in degrees, and matrices with MATLAB's common factor (`1.0e+02 *`).

1.3 About this manual

Chapter 2 gets you from installation to a first solved network in a few minutes. Chapters 3 and 4 explain the window and the editor, and Chapter 5 is the reference for every component and field. Then come per-unit bases (Chapter 6), parametric variables (Chapter 7) and study cases (Chapter 8). Chapter 9 explains each study: what it computes, its options and its output. Chapters 10 to 13 cover reading, editing in bulk, exporting and the command line. Chapter 14 walks through the four EG2100 assignments task by task, Chapter 15 describes the project file, and Chapter 16 lists the messages you may meet and what to do about them. The appendices collect the model equations, the keyboard shortcuts and how the program is verified.

Conventions used in this manual:

- Labels on the screen are set like **Properties**. Menu paths look like File › Export results..., and keys like Ctrl + S.
- Field names, option names and values are set like `p_mw` or `constant_power`. These are the names that bindings, overrides and the project file use.
- File names look like `examples/lf_220kv.psn`, and commands typed in a terminal like `python run_gui.py`.

- The reference tables of Chapters 5 and 9 are generated from the program's own field definitions, so they always match the version this manual describes.

1.4 Scope of version 1.0

PowerLab 1.0 models the balanced three-phase network in per unit, with symmetrical components for unbalanced conditions. It does not yet model breakers and switches as components, three-winding transformers, off-nominal taps or phase shifters, transformer phase shifts in the sequence networks, IEC 60909 short circuits (voltage factors), a distributed slack, or dynamics. Section 16.7 lists the limitations, and `docs/ROADMAP.md` in the repository lists what comes next.

Chapter 2

Getting started

2.1 Requirements and installation

PowerLab is a Python program. It needs:

- Python 3.10 or later (it is tested with the Anaconda distribution, Python 3.12);
- **PySide6** (Qt 6) for the window;
- **numpy**, **scipy** and **networkx** for the engine.

Anaconda already ships numpy, scipy and networkx. PySide6 comes from pip:

```
pip install PySide6 numpy scipy networkx
```

There is nothing to build. Get the repository (`git clone` or a downloaded archive), open a terminal in its root folder, the one that holds `run_gui.py`, and start the program from there. The console tools of Chapter 13 do not need PySide6.

Note

On Windows with Anaconda, Qt 6 can fail to load with *"the specified procedure could not be found"*, because Anaconda ships its own ICU library and that one shadows the library Qt expects. PowerLab works around it: the package `psa.gui` preloads the system ICU when it is imported. `run_gui.py` already imports it first. If you write a script that uses PySide6 together with PowerLab, import `psa.gui` before anything else from PySide6.

2.2 Starting the program

```
python run_gui.py                  # an empty project
python run_gui.py --example        # the validated Assignment S1 network (A = 80)
python run_gui.py my_network.psn   # open a project
```

- `--example` opens `examples/s1_a80.psn` as an *untitled copy*: Ctrl + S asks where to save it and never overwrites the example. File › Load Assignment S1 example does the same from inside the program.

- If the file named on the command line does not exist or is damaged, PowerLab still starts, with an empty project, and tells you why it could not open the file.
- If the project links a variable script (Section 7.5), PowerLab asks before it runs the script. Opening a file never executes code by itself.
- The list of recent files and the window layout are stored per user. On Windows they live in the registry, under `HKCU\Software\PowerLab\PowerLab`.

2.3 A first session in five minutes

This walk-through uses the load flow example that comes with PowerLab, so you can see the whole cycle before drawing anything yourself.

1. Start PowerLab and open `examples/lf_220kv.psn` with File › Open... (Ctrl + O). The canvas shows eight 220 kV buses. The slack generator G1 is at bus 1, an SVC at bus 3, generator G7 at bus 7, constant-power loads at buses 4, 6 and 8, and a capacitor at bus 5.
2. Look at the **Study cases** panel on the right. The project holds three cases: **LFa-LFc) Load flow**, **LFd)**, which raises one load by 10 %, and the **LF variant)** with the shunt rated as a susceptance.
3. Press F5 (Calculate › Run all cases). Each case gets a tab in the **Results** panel at the bottom, and the diagram is annotated with the first one. Every bus shows its voltage, angle and type, every line its power flow and losses, and every generator its P and Q.
4. Read the headline of the first tab: *Converged in 4 iterations. P_loss = 22.1658 MW.* Scroll down for the bus table, the generation table, the branch flows, the power balance, the iteration log and the Jacobian.
5. Click the **LFd)** tab. The annotations switch to that case: the losses rise to 24.3993 MW.
6. In the **Variables** panel, double-click the **Expression** cell of `A`, change `80.0` to `0` and press Enter. The loads that depend on `A` change. The result tabs are marked with an asterisk, and a banner over the canvas says that the results are out of date. Press F5 again: the losses are now 26.4240 MW, the answer printed in the assignment for $A = 0$.
7. Try View › Colour by › Voltage result to colour each bus by its voltage band, and File › Export diagram... to save the annotated drawing as a PNG, SVG or PDF file.

Tip

Edit › Undo (Ctrl + Z) takes back the change of `A` too. Undo covers every change to the network, the layout, the variables, the bases and the study cases.

2.4 The example projects

The folder `examples/` holds one project per assignment. All of them are parametric in the variable `A`, the student's A-number, and are set to $A = 80$. Set it to 0 to reproduce the answers printed in the assignment statements.

File	Contents
`s1_a80.psn`	Assignment S1: a factory fed from a 40 kV grid through two transformers and two cable sections. Cases for the per-unit model, S1a, S1b and S1c.
`s2_grid.psn`	Assignment S2 and S3a–S3c: the meshed 40 kV grid fed from 70 kV, with its zero-sequence data.
`grid_industry.psn`	The S2 grid with the S1 industry connected at bus 7: cases S2a–S2d and S3a–S3e.
`lf_220kv.psn`	Assignment LF: the eight-bus 220 kV network for the load flow, with cases LFa–LFd and the constant-B variant.

Chapter 14 walks through every task of these projects, with the case that answers it and the expected numbers.

Chapter 3

The main window

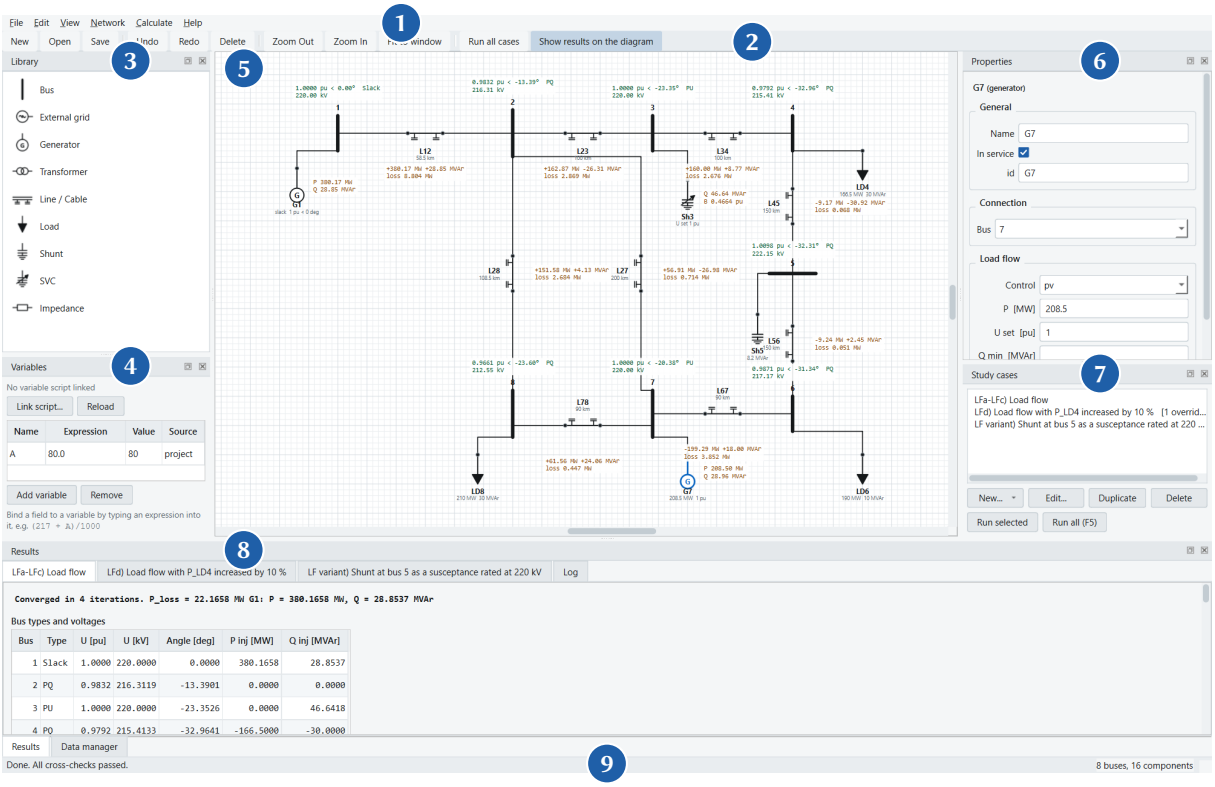


Figure 3.1: The main window with the load flow example after F5. The numbers match the list below.

1. **Menu bar**: **File**, **Edit**, **View**, **Network**, **Calculate** and **Help** (Section 3.1).
2. **Toolbar**: the most used commands (Section 3.2).
3. **Library**: the component palette. Click an entry and then the canvas to place it, or drag it onto the canvas (Section 4.1).
4. **Variables**: the project variables, and those read from a linked script (Chapter 7).
5. **Canvas**: the single-line diagram. With results shown, it carries their annotations (Section 10.4).
6. **Properties**: a form for the selected element or elements, generated from their fields (Section 4.7).
7. **Study cases**: the cases F5 runs, with buttons to create, edit, duplicate, delete and run them (Chapter 8).

8. **Results** and **Data manager**: two tabs of one panel. The first holds a tab per study result plus a **Log** (Chapter 10), the second the component tables (Chapter 11).

9. **Status bar**: messages on the left, and the number of buses and components on the right.

The panels are dock widgets. Drag a panel by its title bar to move it, to stack it with another panel as a tab, or to float it as a separate window. Close one with the × button on its title bar, and bring it back from the **View** menu. PowerLab remembers the layout and the window size between sessions. A layout saved by an older version that lacked one of today's panels is ignored, and the default layout comes back.

3.1 Menus

3.1.1 File

Entry	Shortcut	What it does
New	Ctrl + N	An empty project. Asks first if the current one has unsaved changes.
Open...	Ctrl + O	Open a `.psn` project.
Open recent		The last projects opened.
Load Assignment S1 example		The S1 network as an untitled copy.
Save	Ctrl + S	Save the project; an untitled one asks for a name.
Save As...	Ctrl + ⇧ + S	Save under another name.
Export results...	Ctrl + E	Write the last run's results as text, CSV, LaTeX or JSON (Section 12.1).
Export diagram...	Ctrl + ⇧ + E	Save the diagram as PNG, SVG or PDF (Section 12.2).
Exit	Ctrl + Q	Close PowerLab; asks to save unsaved changes.

3.1.2 Edit

Entry	Shortcut	What it does
Undo, **Redo**	Ctrl + Z, Ctrl + Y	Undo or redo the last change; the entry names it.
Cut, **Copy**, **Paste**	Ctrl + X, Ctrl + C, Ctrl + V	The clipboard, for elements on the canvas (Section 4.9).
Duplicate	Ctrl + D	Copy and paste the selection in one step.
Delete	Del	Delete the selection (Section 4.10).
Select All	Ctrl + A	Select every element.
Find element...	Ctrl + F	Select an element by name or id and centre the diagram on it (Section 4.11).
Rotate 90 deg	Ctrl + Alt + R	Rotate the selected symbols.
Flip bus orientation	Ctrl + Alt + O	Switch the selected buses between vertical and horizontal.

The clipboard shortcuts act on the canvas only when it has the keyboard focus, so that Ctrl + C in a result table copies the table and Ctrl + V in a field pastes text.

3.1.3 View

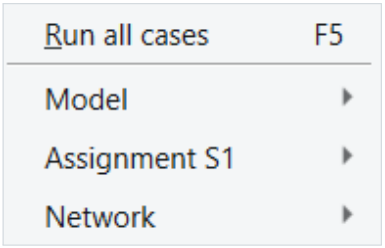
Entry	Shortcut	What it does
Zoom In, **Zoom Out**	Ctrl + [ ] + [ ], Ctrl + -	Zoom the canvas. Ctrl + mouse wheel zooms around the cursor.
Fit to window	Ctrl + 0	Show the whole diagram.
Actual size	Ctrl + 1	One scene unit per pixel.
Show results on the diagram	Ctrl + R	Show or hide the annotations of the selected result (Section 10.4).
Colour by		**Nothing**, **Voltage level** or **Voltage result** (Section 10.5).
Library … **Results**		Show or hide each panel.
Data manager	Ctrl + ⇧ + D	Bring the data manager forward, on the type of the selected element.

3.1.4 Network

Base settings... sets the base power, the frequency, the base reference bus and its base voltage (Chapter 6).

3.1.5 Calculate

Run all cases (F5) runs every study case of the project. A project without cases runs every study once, with its default options. Below it is one submenu per study group: **Model** (per-unit model, admittance matrix, sequence model), **Assignment S1** (S1a–S1c) and **Network** (linear voltages, Thevenin equivalent, short circuit, load flow, unbalanced load).
Choosing a study from these submenus opens the dialog of the study's *own case*, whose id is the study's key. Accept it and the case is saved in the project and run. Section 8.4 explains this in detail.



3.1.6 Help

About PowerLab shows the version and what the program does.

3.2 The toolbar

From left to right: **New**, **Open**, **Save**; **Undo**, **Redo**, **Delete**; **Zoom Out**, **Zoom In**, **Fit to window**; **Run all cases**; and **Show results on the diagram**, a toggle for the annotations. The buttons do the same as the menu entries of the same name.

3.3 When results go stale

Any change to the electrical model makes the results on screen out of date: a field, a connection, an element added, deleted or taken out of service, a variable, the bases. Moving symbols, editing wire bends and dragging labels change only the drawing, so they leave the results alone. PowerLab shows stale results in three ways:

- the result tabs get an asterisk (**LFa-LFc) Load flow** *);
- a banner over the canvas reads *"The model has changed, so the results below are out of date. Press F5 to run the studies again."*;
- the diagram annotations are removed, because they no longer describe the network drawn.

The tables stay readable, so you can compare before and after. Run the cases again to bring everything up to date.

Chapter 4

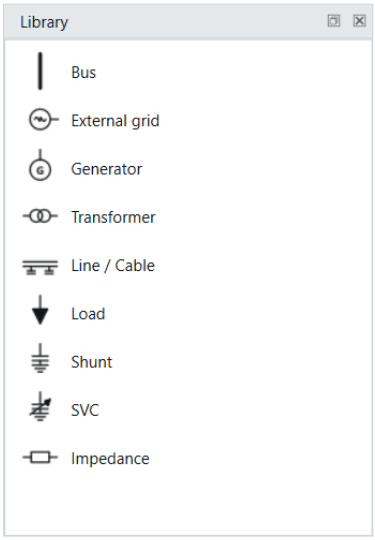
Drawing a network

A network in PowerLab is a set of buses with components connected to them. A component's connections are its *terminals*: one for a load, shunt, generator, SVC or external grid, two for a transformer, line or impedance. What a terminal is connected to is stored in the component itself (its `bus`, `bus_hv`, `bus_from`… fields), so the drawing can never disagree with the electrical model. The wires on the canvas only show those connections.

4.1 Placing components

The **Library** lists every component type (Chapter 5). There are two ways to place one:

- **Click and click.** Click the entry in the library, then click the canvas where the component should go. Hold ⇧ while clicking the canvas to place several in a row. A right-click or Esc drops the tool.
- **Drag and drop.** Drag the entry from the library onto the canvas.



Positions snap to a 10-unit grid. A new component gets an id from its type (`T1`, `LI2`, `LD3`, …) and sensible values to start with: a 10 MVA, 40/12 kV, 7 % transformer, a 1 km cable with $0.16 + j0.08\,\Omega$/km, a 1 MW load at $\cos\varphi = 0.9$ lagging, a 2 MVAr capacitor, a 100 MW PV generator at 1 pu. New buses are named with letters: `a`, `b`, `c`, …

4.1.1 Automatic connection

A component connects itself when it is dropped near buses. Each terminal attaches to the nearest bus within reach (about 95 scene units), and a bus already taken by another terminal of the same component is skipped. So:

- a transformer, line or impedance dropped *between* two buses connects to both;
- a load, shunt, generator or SVC dropped just below a bus hangs from it;
- a component dropped in open space stays unconnected, and you wire it by hand.

4.2 Connecting by hand

Every terminal is drawn as a small circle on its symbol. To connect it:

1. Press the mouse on the terminal and drag. While you drag, every bus the terminal could land on is outlined faintly, and the one under the cursor is outlined strongly.
2. Release on a bus to connect the terminal to it. Release on empty canvas instead, and PowerLab creates a new bus there and connects to it, in one undo step.

Esc or a right-click during the drag cancels it. To disconnect a terminal, set its bus field to empty in **Properties** or in the data manager. A component with an unconnected terminal is allowed: it gives a warning, and the studies leave it out until it is wired.

4.3 Buses

A bus is drawn as a bar, the `busbar` kind, or as a small dot, the `junction` kind, set in its **Kind** field. Junctions suit points inside a feeder, such as the cable joints of S1. A bar is vertical by default.

- **Orientation**: Edit › Flip bus orientation (Ctrl + Alt + O) or the bus's context menu (**Switch vertical / horizontal**).
- **Length**: select the bus and drag the square handle at either end. The opposite end stays put, and the length stays a whole number of grid steps.
- **Where wires land**: a wire meets the bar at the point nearest to the component's terminal, so several branches tap one bar at different points, as in a real single-line diagram.

4.4 Wires

A wire leaves each terminal in the direction the symbol points, and it runs at right angles to its bus. You can shape it with bends:

- **Add a bend**: double-click the wire, or right-click it and choose **Add bend here**.
- **Move a bend**: select the wire, then drag the square handle of the bend.
- **Remove bends**: right-click a bend handle for **Remove this bend**, or the wire for **Remove all bends**.

Bends are stored with the project and move with the drawing. Every bend edit can be undone.

4.5 Symbols and labels

- **Move**: drag a symbol, or a rubber-band selection of several. Wires follow.
- **Rotate**: Edit › Rotate 90 deg (Ctrl + Alt + R) or the context menu. Labels stay upright whatever the rotation.
- **Labels**: every symbol shows its name and a line of key data, for example *58.5 km*, *166.5 MW 30 MVAr* or *81.5 MVA 70/40 kV 8%*. Drag a label to put it somewhere else. The position is saved with the project.
- **Out of service**: an element whose **In service** box is cleared is drawn grey and dashed. No study takes it into account.

4.6 Selecting

Click an element to select it. Drag on empty canvas to draw a rubber band that selects everything it touches. Ctrl + click adds an element to the selection or removes it, and Ctrl + A selects all. The **Properties** panel follows the selection. Several elements of one type are edited together (Section 4.7).

4.7 The property editor

Properties shows a form for the selected element, generated from the element's fields and grouped as in Chapter 5: **General**, **Connection**, **Parameters**, and **Zero sequence** or **Load flow** when the element has such data. The internal **id** is shown read-only at the top. Bindings and overrides refer to elements by this id.

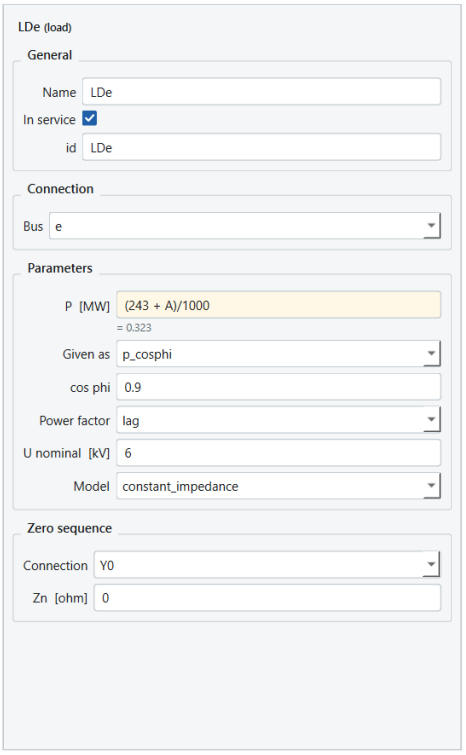


Figure 4.1: A load of S1 whose **P** is bound to the expression `(243 + A)/1000`.

- **Numbers and expressions.** A numeric field takes a number or an expression over the project variables, such as `(243 + A)/1000`. An expression is stored as a *binding*, a link from the field to the expression kept in the project. The field turns amber and shows the value it evaluates to underneath, like a spreadsheet cell (Chapter 7). Clear the field to remove the binding.
- **Validation.** A value outside a field's range (a negative length, a cos φ above 1) is refused. The field turns red and its tooltip says why.
- **Modes.** Fields that depend on a mode appear only when they apply. Set an external grid's **Impedance from** to `ik` and the **Ik** and **cos phi** fields replace **Z Thevenin**.
- **Empty means default.** Some fields accept an empty value with a meaning of its own: a load's **U nominal** empty means "the bus base", and a generator's **Q max** empty means "no limit".
- **Several elements.** With several elements of one type selected, an edit applies to all of them in one undo step. A field whose values differ shows *(several values)*. Names and bus connections are not edited in bulk.
- **Committing.** A field is committed when you press Enter or leave it. Leaving a field without changing it does not create an undo step.

4.8 Context menus

Right-click an element for its menu. It acts on the selection the element belongs to, and it selects the element first if it was not selected.

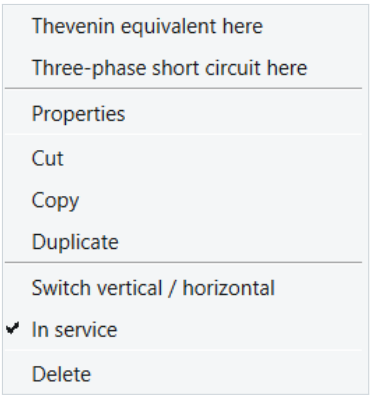


Figure 4.2: The context menu of a bus.

Thevenin equivalent here and **Three-phase short circuit here** (buses only) run that study at this bus at once, with the study's other options at their defaults. The result appears as a new tab. No case is added to the project.

Properties brings the property editor forward on this element.

Cut, Copy, Duplicate are the clipboard commands (Section 4.9).

Rotate 90 deg rotates a symbol. For a bus the entry reads **Switch vertical / horizontal**.

In service is checked while the element is in service. Choose it to take the selection out of service, or to put it back.

Delete deletes the selection.

Right-clicking a wire gives the bend commands described above.

4.9 Clipboard

Ctrl + C copies the selected elements, their layout and their bindings. Ctrl + V pastes them with fresh ids, slightly offset, and selects the pasted elements. Connections among the copied elements are kept. A connection to a bus that was not copied is dropped, so a pasted load whose bus stayed behind arrives unconnected. Ctrl + X cuts and Ctrl + D duplicates (copy and paste in one step). The clipboard holds plain JSON text, so it also works between two PowerLab windows.

4.10 Deleting

Del deletes the selection. Deleting a bus takes along the components that hang from it. PowerLab lists them and asks for confirmation first. Deleting a branch leaves its buses in place. Bindings of deleted elements are removed with them, and undo brings everything back.

4.11 Finding an element

Edit › Find element... (Ctrl + F) opens a list of every bus and component as *name (id, type)*. Type a few letters, choose an entry, or type a name or id (either is matched, case aside) and press Enter. The element is selected, the diagram is centred on it, and its properties are shown. Double-clicking an id in the data manager does the same.

4.12 Undo and redo

Every change to the project is an undoable step: adding, deleting, moving, rotating, connecting, editing fields, wire bends, labels, the bases, the variables and the study cases. Edit › Undo names the step it will undo. Changes of several elements made in one action, such as a multi-selection edit, a paste or deleting a bus with its loads, are one step. Two things are not undoable: reloading the variable script and the annotations toggle.

4.13 Navigating the canvas

- **Zoom**: Ctrl + mouse wheel (around the cursor), Ctrl + + and Ctrl + -. Ctrl + 0 fits the diagram, and Ctrl + 1 is actual size.
- **Pan**: drag with the middle mouse button, or hold Space and drag with the left button. The mouse wheel alone scrolls.

4.14 Tutorial: drawing Assignment S1 by hand

This tutorial rebuilds `examples/s1_a80.psn` from an empty canvas: a 40 kV grid feeding a factory through transformer TI1 (40/12 kV), two cable sections and transformer TI2 (12/6 kV), with loads at buses b and e.

1. File › New. Place five buses in a row from the library, about 240 units apart. They are named a to e. Set the **Kind** of c and d to `junction`.
2. Place an **External grid** to the left of bus a. It hangs from the nearest bus. Leave **U** = 1 pu and **Impedance from** = `z_ohm` with **Z Thevenin** = 0, an ideal source.
3. Drop a **Transformer** between a and b, and set **S rated** = 6 MVA, **U rated HV** = 40 kV, **U rated LV** = 12 kV, **x** = 7 %. Drop another between d and e: 5 MVA, 12/6 kV, 8 %.
4. Drop a **Line / Cable** between b and c and another between c and d. Select both and set, in one edit, $r = 0.16\,\Omega$/km, $x = 0.08\,\Omega$/km and $b_c = 3 \times 10^{-6}$ S/km (type `3e-6`). Give the first one **Parallel lines** = 2.
5. Add the variable `A = 80` in the **Variables** panel. Type `(217 + A)/1000` into the **Length** of both cables.
6. Place a **Load** below bus b with **P** = `(117 - A)/1000`, cos φ = 0.9 lagging and **U nominal** = 12 kV. Place another below bus e with **P** = `(243 + A)/1000` and **U nominal** = 6 kV.
7. Open Network › Base settings... and set the base power to 10 MVA, the base reference bus to a, and the base voltage there to 40 kV. Buses b, c and d get 12 kV and bus e gets 6 kV, by propagation through the transformer ratios.
8. Press F5. Without cases, every study runs with its defaults. The **S1c) Bus voltages** tab gives $U_e = 0.9952\angle -0.5345°$ pu $= 5.9715$ kV, and the **S1b)** tab gives $Z = 22.5232 + j11.1298$ pu.

Save the project with Ctrl + S. The study cases of Chapter 8 then record which buses each study looks at.

Chapter 5

Components

This chapter describes every component type: what it models, how PowerLab turns it into per unit, which studies see it, and every one of its fields. The field tables are generated from the program's own definitions. Each table gives the label shown on screen and, below it, the field's name, which bindings, overrides and the data manager use. Fields that depend on a mode appear in the property editor only when they apply, and the tables say when.

Three fields are common to all components and are not repeated in the text below: **Name**, the label drawn on the diagram (the id stands in when it is empty), **In service**, cleared to make every study ignore the element, and the read-only **id**. The formulas of this chapter are collected, with the zero-sequence rules, in Chapter A.

Note

PowerLab uses one base power S_{base} for the whole network and one base voltage U_{base} per voltage zone (Chapter 6). "Per unit" below always means on those bases. The base impedance at a bus is $Z_{\text{base}} = U_{\text{base}}^2 / S_{\text{base}}$.

5.1 | Bus

A node of the network: a busbar, or a junction drawn as a dot.

Buses are where everything connects. The **Kind** only changes the drawing: a bar for a `busbar`, a dot for a `junction`. Every bus gets its base voltage by propagation from the base reference bus (Chapter 6). A value in **Pinned U base** fixes the base of that bus by hand instead. It also seeds the propagation, and a pinned base that disagrees with a transformer ratio is reported as a conflict.

Field	Unit	Default	Meaning
GENERAL			
Name `name`		–	Label shown on the diagram.
In service `in_service`		on	Out-of-service components are ignored by every study.
PARAMETERS			
Kind `kind`		`busbar`	Junctions are drawn as a small dot instead of a bar. One of `busbar`, `junction`.

Pinned U base u_base_kv	kV	empty	Leave empty to derive the base by propagation from the reference bus through the transformer ratios. At least 0.

5.2 External grid

A voltage source behind a Thevenin impedance: the network beyond the boundary.

The external grid represents the system that feeds the network drawn. It is a voltage $E = U\angle\theta$ behind the impedance Z_{th}. Give the impedance in one of three ways (**Impedance from**):

z_ohm directly, in ohm, referred to the bus base. 0 makes an *ideal* voltage source.

ik from the three-phase fault current I_k at the bus: $|Z_{\text{th}}| = U/(\sqrt{3}\, I_k)$.

sk from the short-circuit power: $|Z_{\text{th}}| = U^2/S_k$.

In the last two, $U = U_{\text{pu}} \cdot U_n$ with U_n the **U nominal** (the bus base when it is empty), and the angle of the impedance follows from the power factor of the fault: $R/|Z| = \cos\varphi$. The S2 grid, for instance, is 70 kV, 25 kA at $\cos\varphi = 0.15$, which gives $Z_{\text{th}} = 0.2425 + j1.5983\,\Omega$.

How the studies see it:

- **Linear solve, Thevenin, faults**: an ideal grid fixes its bus voltage. A grid behind an impedance becomes a Norton pair, $1/Z_{\text{th}}$ to earth plus the current E/Z_{th}. In the negative sequence it is Z_{th} without a source voltage.
- **Admittance matrix**: depending on the study's **Sources** option, the grid is an extra internal node behind Z_{th}, a shunt $1/Z_{\text{th}}$ to earth, or left out (Section 9.2).
- **Load flow**: with **Slack** checked, the grid is the slack bus at its own terminal, holding U and θ. Its impedance is short-circuit data and does not enter the load flow.
- **Zero sequence** (**Z0 from**): none, an ungrounded system with no zero-sequence path; z0_ohm, Z_0 given in ohm; ik1, from the single line-to-ground fault current, $Z_0 = 3U_{\text{ph}}/I_{k1}\angle\arccos(\cos\varphi_0) - 2Z_1$; or z0_z1, a multiple of Z_1.

Field	Unit	Default	Meaning
GENERAL			
Name name		–	Label shown on the diagram.
In service in_service		on	Out-of-service components are ignored by every study.
Bus bus		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.
PARAMETERS			
U u_pu	pu	1	Source voltage; with Ik or Sk also the pre-fault voltage the fault level refers to. At least 0.
Angle theta_deg	deg	0	

Field	Unit	Default	Meaning
Impedance from `z_mode`		`z_ohm`	z_ohm: give Z directly. ik: three-phase fault current. sk: short-circuit power. One of `z_ohm`, `ik`, `sk`.
Z Thevenin `z_th_ohm`	ohm	0	0 means an ideal voltage source. Referred to the bus base. Shown when **Impedance from** is `z_ohm`.
Ik (three-phase) `ik_ka`	kA	0	At least 0. Shown when **Impedance from** is `ik`.
Sk (three-phase) `sk_mva`	MVA	0	At least 0. Shown when **Impedance from** is `sk`.
cos phi of the fault `cos_phi`		0.1	Lagging power factor of the fault current: R/|Z|. From 0 to 1. Shown when **Impedance from** is `ik` or `sk`.
U nominal `un_kv`	kV	empty	Voltage the fault level is quoted at. Empty means the bus base. At least 0. Shown when **Impedance from** is `ik` or `sk`.
Slack `is_slack`		on	Fixes the voltage angle reference for the network.
ZERO SEQUENCE			
Z0 from `z0_mode`		`none`	none: no zero-sequence path (ungrounded). z0_ohm: give Z0. ik1: single line-to-ground fault current. z0_z1: Z0/Z1. One of `none`, `z0_ohm`, `ik1`, `z0_z1`.
Z0 `z0_ohm`	ohm	0	Shown when **Z0 from** is `z0_ohm`.
Ik1 (line-to-ground) `ik1_ka`	kA	0	At least 0. Shown when **Z0 from** is `ik1`.
cos phi of that fault `cos_phi0`		0.1	From 0 to 1. Shown when **Z0 from** is `ik1`.
Z0 / Z1 `z0_z1`		1	At least 0. Shown when **Z0 from** is `z0_z1`.

5.3 Generator

A synchronous generator, for the load flow.

The generator exists for the load flow. Its **Control** sets what it holds:

`pv` its active power **P** and its bus voltage **U set**: the bus becomes a PU bus. **Q min** and **Q max** limit its reactive power when the load flow is asked to enforce limits; empty means no limit.

`slack` the voltage **U set** and its angle **Angle**: the bus becomes the slack bus, which balances the network.

`pq` a fixed **P** and **Q**: the bus stays a PQ bus.

The linear, Thevenin and fault studies do not see generators. They work with external grids, so give a network an external grid when you want those studies too. Several voltage-controlling devices on one bus (two generators, or a generator and an SVC) must ask for the same voltage, otherwise the load flow refuses to run. They share the bus's reactive power equally.

Field	Unit	Default	Meaning
GENERAL			
Name `name`		–	Label shown on the diagram.
In service `in_service`		on	Out-of-service components are ignored by every study.
Bus `bus`		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.
LOAD FLOW			
Control `control`		`pv`	pv: holds P and the bus voltage. slack: holds the voltage and its angle and balances the network. pq: holds P and Q. One of `pv`, `slack`, `pq`.
P `p_mw`	MW	0	Shown when **Control** is `pv` or `pq`.
Q `q_mvar`	MVAr	0	Shown when **Control** is `pq`.
U set `u_set_pu`	pu	1	At least 0. Shown when **Control** is `pv` or `slack`.
Angle `theta_deg`	deg	0	Shown when **Control** is `slack`.
Q min `q_min_mvar`	MVAr	empty	Empty: no limit. Shown when **Control** is `pv`.
Q max `q_max_mvar`	MVAr	empty	Empty: no limit. Shown when **Control** is `pv`.

5.4 Transformer

A two-winding transformer: a series impedance, with the ratio absorbed by the base change.

The transformer is its short-circuit impedance, r and x in percent on its own rating. In per unit on the network bases,

$$z = \frac{r + jx}{100} \cdot \frac{S_{\text{base}}}{S_n} \cdot \left(\frac{U_n}{U_{\text{base}}} \right)^2 ,$$

where U_n is the rated voltage of the side chosen in **Refer Z to** and U_{base} the base of the bus on that side. There is no magnetising branch. The turns ratio does not appear because the bases on its two sides are in the rated ratio (Chapter 6). When pinned bases make the two sides disagree, the per-unit table reports the value the other side would give.

Caution

Tap ratio is stored with the project but not modelled yet: an off-nominal tap gives a warning, the transformer is treated as a plain series impedance, and the $\det F = 1$ check of the two-port study no longer applies.

Zero sequence. The **Vector group** decides the zero-sequence path, with the HV winding first: `YN` is a grounded star, `Y` an ungrounded star, `D` a delta. With Z_0 from **x0** and **r0** (equal to x and r when empty), and Z_n the neutral impedances, appearing as $3Z_n$:

- `YNyn`: a series branch $Z_0 + 3Z_{n,\mathrm{HV}} + 3Z_{n,\mathrm{LV}}$;
- `YNd`: $Z_0 + 3Z_{n,\mathrm{HV}}$ from the HV bus to earth, the LV side open;
- `Dyn`: $Z_0 + 3Z_{n,\mathrm{LV}}$ from the LV bus to earth, the HV side open;
- every other group: open in the zero sequence.

Phase shifts (Yd11, Dyn5…) are not modelled, as in the course.

Field	Unit	Default	Meaning
GENERAL			
Name `name`		–	Label shown on the diagram.
In service `in_service`		on	Out-of-service components are ignored by every study.
HV bus `bus_hv`		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.
LV bus `bus_lv`		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.
PARAMETERS			
S rated `sn_mva`	MVA	1	At least 0.
U rated HV `u1n_kv`	kV	1	At least 0.
U rated LV `u2n_kv`	kV	1	At least 0.
x `x_pct`	%	0	
r `r_pct`	%	0	
Tap ratio `tap`		1	Off-nominal ratio. Must be 1.0 for the S1 studies. At least 0.
Refer Z to `referred_side`		`lv`	Which side's rated voltage and base are used for the per-unit conversion. One of `lv`, `hv`.
ZERO SEQUENCE			
Vector group `vector_group`		`YNyn`	HV winding first: YN grounded star, Y star, D delta. Decides the zero-sequence path; phase shifts are not modelled. One of `YNyn`, `YNy`, `Yyn`, `Yy`, `YNd`, `Yd`, `Dyn`, `Dy`, `Dd`.
Zn HV `zn_hv_ohm`	ohm	0	Neutral-to-earth impedance of a grounded HV star (0 = solid).
Zn LV `zn_lv_ohm`	ohm	0	Neutral-to-earth impedance of a grounded LV star (0 = solid).
x0 `x0_pct`	%	empty	Empty means the same as x.
r0 `r0_pct`	%	empty	Empty means the same as r.

5.5 Line / Cable

A PI section, with n identical lines in parallel.

A line is a PI equivalent built from its per-kilometre data and its length ℓ. With n lines in parallel,

$$z = \frac{(r + jx)\,\ell}{n\,Z_{\text{base}}}, \qquad \frac{y_{\text{sh}}}{2} = \frac{n\,(g + jb_c)\,\ell}{2}\,Z_{\text{base}} \quad \text{at each end.}$$

b_c is the *total* charging susceptance per kilometre of *one* line. Half of it goes to each end of the PI. A line must join two buses with the same base voltage. Use a transformer to change voltage level.

Zero sequence. **Z0 data** must be set before any zero-sequence study runs: `factors` multiplies r, x (and g, b_c by **b0 / b**), and `absolute` gives r_0, x_0, b_0 per kilometre. While it is `unset`, the Thevenin study leaves the zero sequence out with a warning, and the studies that need it (single line-to-ground and double line-to-ground faults, unbalanced loads, the zero-sequence admittance matrix) refuse to run and name the line.

Field	Unit	Default	Meaning
GENERAL			
Name `name`		–	Label shown on the diagram.
In service `in_service`		on	Out-of-service components are ignored by every study.
From bus `bus_from`		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.
To bus `bus_to`		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.
PARAMETERS			
r `r_ohm_per_km`	ohm/km	0	
x `x_ohm_per_km`	ohm/km	0	
b_c `b_s_per_km`	S/km	0	Total charging susceptance per km of ONE line.
g `g_s_per_km`	S/km	0	
Length `length_km`	km	1	At least 0.
Parallel lines `n_parallel`		1	At least 1.
ZERO SEQUENCE			
Z0 data `z0_mode`		`unset`	unset: zero-sequence studies refuse to run. factors: multiples of r, x, b. absolute: per-km values. One of `unset`, `factors`, `absolute`.
r0 / r `r0_factor`		3	At least 0. Shown when **Z0 data** is `factors`.
x0 / x `x0_factor`		3	At least 0. Shown when **Z0 data** is `factors`.

Field	Unit	Default	Meaning
b0 / b `b0_factor`		0.5	At least 0. Shown when **Z0 data** is `factors`.
r0 `r0_ohm_per_km`	ohm/km	0	At least 0. Shown when **Z0 data** is `absolute`.
x0 `x0_ohm_per_km`	ohm/km	0	Shown when **Z0 data** is `absolute`.
b0 `b0_s_per_km`	S/km	0	Shown when **Z0 data** is `absolute`.

5.6 Load

A load given by its power at a nominal voltage.

Give the load's power as **P** with **cos phi** and **Power factor** (`lag` draws inductive reactive power, `lead` capacitive), or as **P** and **Q** directly (**Given as** = `p_q`, positive Q inductive). Its **Model** decides how the studies see it:

`constant_impedance` the load is the admittance that draws its power at **U nominal** (the bus base when empty): $y = \dfrac{S^*}{S_{\text{base}}} \left(\dfrac{U_{\text{base}}}{U_n} \right)^2$. A load quoted at 12 kV on a bus whose base is 12.6 kV therefore draws more than its nominal power at 1 pu, as a real impedance load would.

`constant_power` in the load flow, the load is a PQ injection that draws exactly its power whatever the voltage. The linear, Thevenin and fault studies cannot represent constant power: they keep the constant impedance and say so in a warning.

Zero sequence and unbalance. **Connection** `Y0` is a grounded star, whose zero-sequence admittance is $1/(1/y + 3Z_n)$. `Y` and `D` are open in the zero sequence. The unbalanced load study (Section 9.10) uses the connection to build the load's phase admittance matrix.

Field	Unit	Default	Meaning
GENERAL			
Name `name`		–	Label shown on the diagram.
In service `in_service`		on	Out-of-service components are ignored by every study.
Bus `bus`		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.
PARAMETERS			
P `p_mw`	MW	0	
Given as `spec`		`p_cosphi`	P and cos phi, or P and Q. One of `p_cosphi`, `p_q`.
cos phi `cos_phi`		1	From 0 to 1. Shown when **Given as** is `p_cosphi`.

Field	Unit	Default	Meaning
Power factor `pf_sign`		`lag`	One of `lag`, `lead`. Shown when **Given as** is `p_cosphi`.
Q `q_mvar`	MVAr	0	Positive is inductive (the load absorbs reactive power). Shown when **Given as** is `p_q`.
U nominal `u_nominal_kv`	kV	empty	Voltage at which the load consumes P. Empty means the bus base. At least 0.
Model `model`		`constant_impedance`	The load flow honours constant_power (a PQ injection); the linear, Thevenin and fault studies treat every load as a constant impedance at its nominal voltage. One of `constant_impedance`, `constant_power`.
ZERO SEQUENCE			
Connection `connection`		`Y0`	Y0: grounded star (a zero-sequence path). Y and D: none. One of `Y0`, `Y`, `D`.
Zn `zn_ohm`	ohm	0	Neutral-to-earth impedance (0 = solidly grounded). Shown when **Connection** is `Y0`.

5.7 Shunt

A shunt capacitor ($Q > 0$) or reactor ($Q < 0$).

The shunt is rated by the reactive power Q it delivers at **U nominal**. Positive Q is capacitive, the same sign as line charging. Use a negative value for a reactor; the symbol then changes to a coil. As an admittance, $y = j\,\dfrac{Q}{S_{\text{base}}}\left(\dfrac{U_{\text{base}}}{U_n}\right)^2$.

In the load flow, **Model** chooses between `constant_b`, the admittance above, which delivers $B\,U^2$, and `constant_q`, an injection of Q whatever the voltage. The two readings are the base case and the variant of Assignment LF. Every other study uses the admittance. **Connection** works as for loads.

Field	Unit	Default	Meaning
GENERAL			
Name `name`		–	Label shown on the diagram.
In service `in_service`		on	Out-of-service components are ignored by every study.
Bus `bus`		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.
PARAMETERS			
Q `q_mvar`	MVAr	0	Positive is capacitive, the same sign as line charging. Use a negative value for a reactor.
U nominal `u_nominal_kv`	kV	empty	At least 0.
LOAD FLOW			
Model `model`		`constant_b`	constant_b: Q is rated at U nominal and scales with U^2. constant_q: the load flow injects Q whatever the voltage. One of `constant_b`, `constant_q`.
ZERO SEQUENCE			

Field	Unit	Default	Meaning
Connection `connection`		`Y0`	Y0: grounded star (a zero-sequence path). Y and D: none. One of `Y0`, `Y`, `D`.

5.8 SVC

A voltage-controlled shunt: a static var compensator or a synchronous condenser.

In the load flow, the SVC holds its bus at **U set** with whatever reactive power that takes, so its bus is a PU bus with $P = 0$. The results report that Q, and the susceptance it amounts to, $B = Q/U^2$, in per unit and in siemens. This is how Assignment LF sizes the controllable shunt at bus 3. **Q min** and **Q max** limit it when the load flow enforces limits. The other studies do not see the SVC.

Field	Unit	Default	Meaning
GENERAL			
Name `name`		–	Label shown on the diagram.
In service `in_service`		on	Out-of-service components are ignored by every study.
Bus `bus`		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.
LOAD FLOW			
U set `u_set_pu`	pu	1	At least 0.
Q min `q_min_mvar`	MVAr	empty	Empty: no limit (negative is inductive).
Q max `q_max_mvar`	MVAr	empty	Empty: no limit.

5.9 Impedance

A raw $R + jX$, in series between two buses or to earth.

A plain impedance, for whatever the other components do not cover: a series reactor, a fault path, an equivalent. Give R and X in ohm, referred to the base of the **From bus**, or check **Values in pu**. With **To bus** empty, the impedance goes from its bus to earth. It is the same in every sequence.

Field	Unit	Default	Meaning
GENERAL			
Name `name`		–	Label shown on the diagram.
In service `in_service`		on	Out-of-service components are ignored by every study.
From bus `bus_from`		–	The bus this terminal is connected to: wire it on the canvas, or pick the bus here.

Field	Unit	Default	Meaning
To bus `bus_to`		empty	Leave empty for a shunt to ground.
PARAMETERS			
R `r`	ohm / pu	0	
X `x`	ohm / pu	0	
Values in pu `in_pu`		off	

Chapter 6

Per-unit bases

Every study works in per unit. PowerLab follows the convention of the course:

- one base power S_{base} for the whole network;
- one base voltage U_{base} per voltage zone, the line-to-line voltage;
- from those, at every bus,

$$Z_{\text{base}} = \frac{U_{\text{base}}^2}{S_{\text{base}}}, \qquad I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}\,U_{\text{base}}}.$$

6.1 Base settings

Network › Base settings... sets four values:

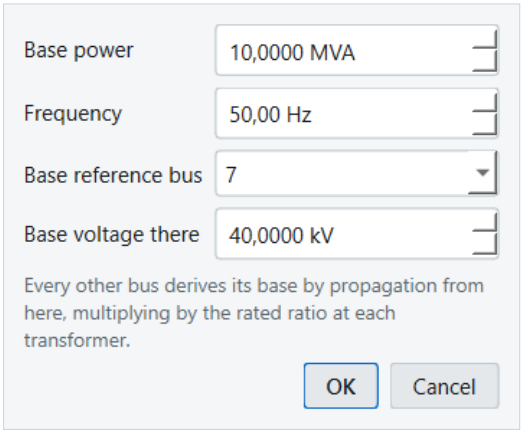
Base power S_{base} in MVA.

Frequency in Hz, kept with the project.

Base reference bus the bus where you know the base voltage.

Base voltage there its U_{base} in kV.

The change is one undoable step, and it makes the results stale.



6.2 How bases propagate

From the reference bus, PowerLab walks through the network. Crossing a line or an impedance keeps the base. Crossing a transformer multiplies it by the transformer's *rated* ratio, U_{2n}/U_{1n} from HV to LV. For S1, 40 kV at bus a becomes 12 kV beyond TI1 (40/12 kV) and 6 kV beyond TI2 (12/6 kV). With bases in the rated ratio, a transformer in per unit is a plain series impedance: its ratio disappears into the base change.

- **Bases follow the nameplate, not the switching state.** The walk crosses every branch, in service or not, so taking a branch out of service does not strip a bus of its base. A case that disconnects the industry still prints the industry's buses with their bases.
- **Pinned bases.** A value in a bus's **Pinned U base** sets that bus's base by hand and seeds the walk from there as well. Pin a base when a part of the network hangs from the rest only through elements that do not carry a base, or to follow a textbook that chooses bases other than the rated ratios.
- **Voltage zones.** Buses with the same base form a zone. The **Per-unit model** study lists the zones and the bases of every bus (Section 9.1).

6.3 When bases cannot be established

Conflict. Around a loop of transformers whose ratios do not multiply to one, a bus is reached twice with different bases. PowerLab does not choose one silently. It reports *"Bus 'x' already has U_base = 12 kV but is also reached with 12.6 kV through 'T3'. Check the transformer ratios around this loop."*, naming the bus and the transformer, and the studies do not run until it is resolved. Fix the ratio, or pin the base you want.

Unreachable bus. A bus that no path connects to the reference bus has no base: *"Bus 'x' cannot be reached from the base reference bus, so it has no per-unit base."* Connect it, pin its base, or take it out of service. A bus that only a de-energized part of the network reaches is left out of the network studies with an information message instead of an error.

6.4 How each component uses the bases

- A transformer's impedance is converted on the base of the side chosen in its **Refer Z to** field, with the factor $(U_n/U_{\text{base}})^2$, which is 1 when the bases follow the rated voltages.
- Loads and shunts are rated at their own **U nominal**. Their admittance carries the factor $(U_{\text{base}}/U_n)^2$, shown in the per-unit table so it can be checked.
- Lines, impedances in ohm and external grids use the base of their own bus. A line must join two buses of the same zone.

The **(Ub/Un)^2** column of the per-unit model shows every one of these factors.

Chapter 7

Parametric variables

Assignment data is often algebraic: a cable *217 + A* metres long, a load of *(117 − A)* kW. PowerLab lets you write exactly that. Define `A` once, write each field as an expression in `A`, and change `A` to re-evaluate the whole network. This is how the example projects reproduce both the $A = 80$ solution and the $A = 0$ answer printed in the statements.

7.1 Three layers

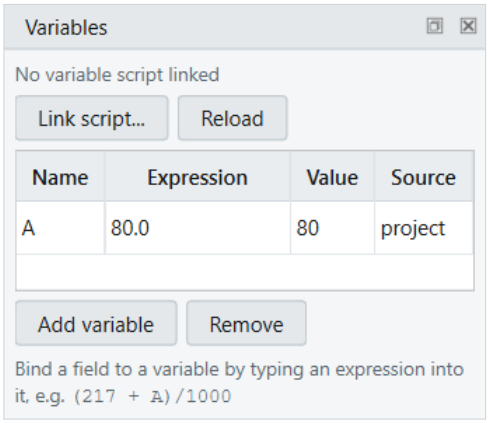
Values are evaluated in this order, each layer able to use the names of the layers before it:

1. **Script variables**, the numeric globals of a Python file you link to the project (Section 7.5).
2. **Project variables**, ordered *name = expression* pairs edited in the **Variables** panel. A variable may use the variables above it and the script's.
3. **Bindings**, *COMPONENT.field = expression*. They are written into the network's fields whenever a variable changes.

The network always holds evaluated numbers. The expressions live beside it, in the project file's `parameters` section, so the studies never see an expression.

7.2 The Variables panel

The table lists every variable with its **Name**, **Expression**, evaluated **Value** and **Source**: `project` for those defined here, `script` for those read from the linked file.



- **Add variable** adds a row. Type its name and expression in the table.
- Edit a project variable's name or expression by double-clicking its cell. Script variables are read-only here; edit them in the script.
- **Remove** deletes the selected project variable.
- A variable whose expression fails shows the error as the tooltip of its cell.

Every change re-evaluates the bindings, makes the results stale and can be undone. Undo puts the network back to its previous values too.

7.3 Binding a field

Type an expression instead of a number into any parametric numeric field, in the property editor or in the data manager. The field turns amber and shows its value underneath (Section 4.7). Clear the field to remove the binding. The field then keeps its last value, or becomes empty if empty is a value it accepts. Bound cells are tinted in the data manager too. Fields marked *Numbers only* in the reference tables, such as the number of parallel lines, do not take expressions.

7.4 Expressions

An expression may use:

- numbers, including complex ones: `1.6+0.3j`. Complex fields also accept MATLAB's `1.6+0.3i`;
- the operators `+ - * / // % **` and parentheses;
- the variables defined before it;
- the functions `sqrt`, `sin`, `cos`, `tan`, `asin`, `acos`, `atan`, `atan2`, `exp`, `log`, `log10`, `hypot`, `degrees` (also `deg`), `radians` (also `rad`), `abs`, `min`, `max`, `round`, `float` and `int`;
- the constants `pi` and `e`.

Attribute access, indexing, keyword arguments and lambdas are not allowed: expressions are evaluated by a whitelist, never by Python's `eval`. A failing expression says why, for example *"Unknown name 'B'. Define it as a variable or load it from your script."* or *"Division by zero."*

For example, the external grid of S2 (70 kV, 25 kA at $\cos\varphi = 0.15$) can be written directly as its Thevenin impedance:

```
70/(sqrt(3)*25)*(0.15 + 1j*sqrt(1 - 0.15**2))      # = 0.2425 + 1.5983j ohm
```

The example projects use the **Impedance from** = `ik` mode instead, which says the same with less algebra.

7.5 A variable script

For many variables, or variables computed with some logic, write them in a Python file of your own, such as `examples/variables_S1.py`, and link it with **Link script...** in the **Variables** panel. PowerLab runs the file and keeps its public numeric globals: numbers whose names do not start with an underscore.

Caution

The script is ordinary Python and runs with your privileges. PowerLab asks for confirmation when you link one, and again every time you open a project that links one. If you decline, the values saved in the project are used. **Reload** runs the script again when you have edited it. Only link files you trust.

The script's path is saved in the project. A relative path is read from the project's folder, which keeps a project and its script movable together. Linking a script is an undoable step; reloading it is not.

7.6 Variables and study cases

Study cases can override fields with expressions too, and those expressions see the project variables plus `value`, the field's own value (Section 8.2.2). A case that sets `LD4.p_mw` to `1.1*value` raises that load by 10 % whatever `A` is. On the command line, `-set A=0` changes a variable before solving (Chapter 13).

Chapter 8

Study cases

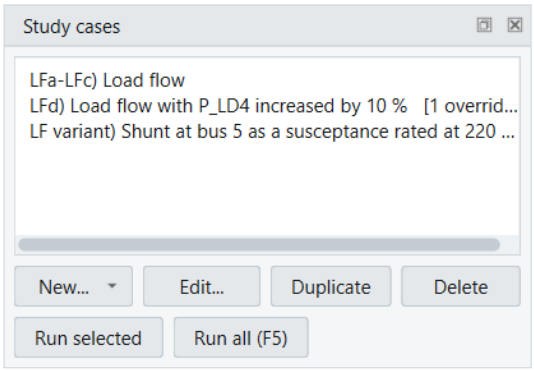
A *study case* is one question asked of the network:

- a **study**, such as the Thevenin equivalent or the load flow;
- its **options**, such as the bus to look at or the fault type;
- optionally, a **network variant**: elements taken *out of service* and fields *overridden* by expressions.

Cases are saved with the project, so one project holds every question about a network side by side: “industry disconnected”, “load at 50 %”, “+10 % load at bus 4”. The variant is applied to a copy of the network when the case runs. The network you drew never changes.

8.1 The Study cases panel

The panel lists the project’s cases in the order F5 runs them. A case with a variant says so after its title, for example *[1 override(s)]* or *[1 out of service]*. Hover a case to see its id and study.



New... opens a menu of every study, grouped as in the **Calculate** menu. Pick one to create a case and edit it.

Edit... edits the selected case. Double-clicking a case does the same.

Duplicate inserts a copy below it, with a new id and *(copy)* added to its title.

Delete removes it.

Run selected runs the selected cases. Hold Ctrl or ⇧ to select several.

Run all (F5) runs every case.

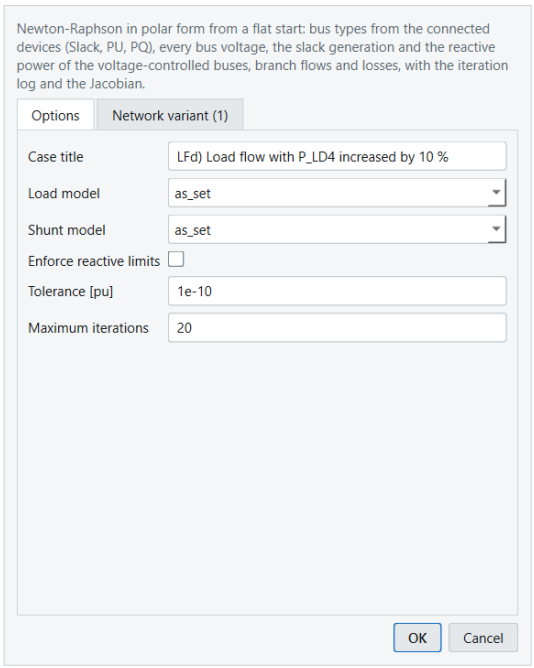
Every change to the list (new, edit, duplicate, delete) is undoable. A project without cases is valid. F5 then runs every study once with its default options, which is a quick way to see everything a new network gives.

8.2 The case dialog

The dialog shows the study's description, plus a note for studies that need one, and two tabs.

8.2.1 Options

Case title names the case in the panel and in its result tab. Left empty, the study's title stands in.
The other rows are the study's options, generated from its definition and described for each study in Chapter 9. Bus options list the network's buses. Element options list the elements of the right type. Choice options offer their values. Options that only apply in some situations appear when they do: the ground impedance of a fault, for instance, appears for a double line-to-ground fault only.
Some options depend on others. In S1a, the list of shunts that may be included at the ports changes with the port buses chosen (Section 9.4).

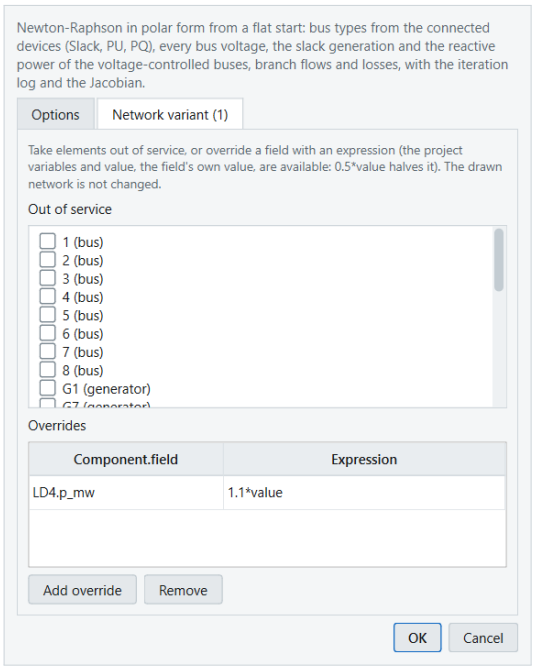


8.2.2 Network variant

Out of service. Every bus and component is listed with a check box. A checked element is out of service in this case only. Assignment S2 uses this to study the grid with the S1 industry disconnected (TI1 out of service in `grid_industry.psn`).
Overrides. Each row sets a field, written `COMPONENT_ID.field` (for example `LD4.p_mw`), to an expression. The expression may use the project variables and `value`, the field's current value:

- `1.1*value` raises the field by 10 % (Assignment LFd);
- `0.5*value` halves it (Assignment S2d);
- `A/1000` sets it from a variable.

Add override adds a row, and **Remove** deletes the selected rows. Field names are the ones in the reference tables of Chapter 5, and in the data manager's column tooltips.



When you press **OK**, PowerLab checks every override: the component must exist, the field must be one of its fields, and the expression must evaluate. A mistake is shown in red under the tabs and the dialog stays open. The tab's title counts the variant's entries, for example **Network variant (1)**.

Note

Overrides are applied in the order shown, on a copy of the network taken after the project's own bindings were evaluated. When a later edit deletes or renames an element that a case refers to, the case runs anyway and its result starts with a warning naming the stale entry.

8.3 Running cases

Cases run in three ways:

- F5, Calculate › Run all cases or **Run all (F5)**: every case of the project, in order;
- **Run selected**: the selected cases;
- a study from the **Calculate** menu, which runs that study's own case (Section 8.4), or a bus's context menu, which runs a Thevenin equivalent or a three-phase fault at that bus without saving a case.

Before running, PowerLab validates the network. An error, such as a missing base reference bus or a branch that joins a bus to itself, stops the run with a list of what to fix (Chapter 16). A warning, such as an unconnected component, does not stop it. Each case then gets a result tab. A case that cannot run (S1a with ports that are not connected, a Thevenin equivalent in a network without a source) gets a tab with the reason, and the other cases are still shown. Only when every case fails does PowerLab show the error in a message box instead.

8.4 A study's own case

Choosing a study from the **Calculate** menu does not run it blindly. It opens the dialog of the study's *own case*, the case whose id is the study's key (`thevenin`, `loadflow`…), creating it the first time with the study's defaults. Accept the dialog and the case is saved in the project and run. The next time you choose the study, the dialog opens with the options you left. These cases appear in the **Study cases** panel like any other, so F5 runs them too. Studies without options (the per-unit model and the sequence model) run straight away.

8.5 Default options

A study's options start from defaults chosen for the network at hand:

- the bus options of the Thevenin and short-circuit studies default to the base reference bus, unless an ideal source sits there, in which case the bus furthest from it is used;
- S1a's ports default to the reference bus and the bus furthest from it;
- the linear voltages report the bus furthest from the reference bus;

- the unbalanced load defaults to the last load of the network, with its phase factors at 1, and reports the voltages at its own bus.

An option left empty in a case falls back to its default when the case runs.

Chapter 9

Studies

This chapter describes the eleven studies: what each one computes and how, its options, what its result tab shows, how it checks itself, and what it draws on the diagram. The option tables are generated from the studies' definitions.

Study	Menu group	In short
Per-unit model (`model`)	Model	Bases and every component in per unit.
Admittance matrix (`ybus`)	Model	Y of the positive, negative or zero sequence.
Sequence model (`seqmodel`)	Model	What path each element gives in each sequence.
S1a) Two-port matrix (`s1a`)	Assignment S1	ABCD matrix between two buses.
S1b) Equivalent impedance (`s1b`)	Assignment S1	$Z = U/I$ into the passive network.
S1c) Bus voltages (`s1c`)	Assignment S1	Linear solve with the S1 phasor chain.
Bus voltages (`linear`)	Network	Linear solve of any network.
Thevenin equivalent (`thevenin`)	Network	U_{th} and $Z_{\mathrm{th},1/2/0}$ at a bus.
Short circuit (`short_circuit`)	Network	3ph, SLG, LL or LLG fault.
Load flow (`loadflow`)	Network	Newton–Raphson load flow.
Unbalanced load (`unbalanced_load`)	Network	A load with different power per phase.

Which part of the network a study sees. Out-of-service elements are always ignored. The network studies (admittance matrix, linear voltages, Thevenin, short circuit, load flow, unbalanced load, sequence model) also keep only the *energized* part of the network: the islands that contain an external grid or a generator. A case that disconnects part of the network simply sees the rest, and the disconnected buses are reported as information. The S1 studies analyse the whole network as drawn.

Loads in the linear studies. Every study except the load flow treats every load as a constant impedance at its nominal voltage, which keeps the problem linear and exact in one solve. A load set to `constant_power` gets a warning in those studies.

9.1 Per-unit model

The starting point of every hand calculation: the bases and each component in per unit, with the factors that took it there.

This study has no options.

Result.

- **Base values**: U_{base}, Z_{base} and I_{base} of every bus, with S_{base} in the note;
- **Voltage zones**: the buses that share each base voltage;
- **Component values in per unit**: for each component, its Z, its shunt admittance (the PI half-shunt for lines, the total admittance for loads and shunts), its power in MVA, its nominal voltage, the factor $(U_{\text{base}}/U_n)^2$ and a note, such as *2 parallel line(s), 0.297 km* or *capacitive*.

9.2 Admittance matrix (Y-bus)

The bus admittance matrix of one sequence network of the energized network, the way the course writes it: S2a (positive sequence, source as an internal node, 9×9) and S3a (zero sequence, source grounded, 8×8).

Option	Kind	Meaning
Sequence `sequence`	choice	One of `positive`, `negative`, `zero`.
Sources `sources`	choice	internal_node: each source behind its impedance becomes an extra node (the S2a form). grounded: its impedance goes to ground at the bus (the S3a form). exclude: leave sources out. One of `internal_node`, `grounded`, `exclude`.

Sources. A source with zero impedance never enters the matrix, because as an admittance it would be a short circuit. A source behind an impedance enters as follows:

`internal_node` as an extra node *(internal)* behind a series branch Z_{th}, placed after the buses;

`grounded` as a shunt $1/Z_{\text{th}}$ to earth at its bus (the Norton form);

`exclude` not at all: the passive network.

Result. The matrix, with MATLAB's common factor in the title when there is one (*1.0e+02 **). Below it are the **Nodes** table, which lists each row with its node, id and U_{base}, and a **Checks** table with the symmetry residual $\max|Y - Y^T|$ and the source treatment. Right-click the matrix for **Copy as MATLAB** (Section 10.3.1).

9.3 Sequence model

For each element, the path it gives in the positive and the zero sequence, with its impedance or admittance in per unit and a note. The note says, for example, that a `Dyn` transformer is a shunt on its star side, that a delta load is open, or that a grid's Z_0 comes from $3U/I_{k1} - 2Z_1$. The negative sequence equals the positive one without source voltages, so it is not printed separately. When a line has no zero-sequence data, the zero-sequence table is replaced by a warning that names it.

This study has no options.

9.4 S1a) Two-port matrix

The transmission (ABCD) matrix between two buses, with the course's convention

$$\begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = F \begin{bmatrix} U_2 \\ I_2 \end{bmatrix}, \qquad F = \begin{bmatrix} A & B \\ C & D \end{bmatrix},$$

port 1 on the source side and port 2 on the load side. Swapping the ports gives a different matrix.

Option	Kind	Meaning
From bus (port 1, source side) `bus_i`	bus	Any bus of the network.
To bus (port 2, load side) `bus_j`	bus	Any bus of the network.
Shunts at the ports `include_at_ports`	elements	A load or source sitting on a port bus is the termination, not part of the two-port, so it is excluded. Tick one to include it anyway. Line charging is always kept.

The termination rule. The two-port is the network *between* the ports. A load or source sitting on a port bus is the termination, not part of the two-port, so it is excluded. The same holds for whatever hangs off a port without leading to the other one, such as the rest of a feeder beyond port 2. The dialog lists the shunts on the port buses, and you can tick one to include it anyway. An ideal source on a port can never be included, because it would short-circuit the port. Line charging is always kept, because it belongs to the line.

Two methods. The matrix is computed twice. First by cascading the element matrices along the path,

$$F_{\text{series}}(z) = \begin{bmatrix} 1 & z \\ 0 & 1 \end{bmatrix}, \quad F_{\text{shunt}}(y) = \begin{bmatrix} 1 & 0 \\ y & 1 \end{bmatrix}, \quad F_\pi = F_{\text{shunt}}(y_{\text{sh}})\, F_{\text{series}}(z)\, F_{\text{shunt}}(y_{\text{sh}}),$$

and then by Kron reduction of the Y-bus onto the two ports. The relative residual between the two is reported. It is a real check: the line charging lives inside F_π in one method and on the diagonal of Y in the other, so counting it twice would show up as a residual of about 10^{-5}. In a meshed network the path is not unique and the cascade does not exist; the result then comes from the Y-bus alone, with a warning.

Result. The convention, the elements excluded at the ports and why, each element's matrix in cascade order, F itself, and the checks: $\det F$ (equal to 1 for a reciprocal network), $|\det F - 1|$, the cascade-versus-Y-bus residual and the cascade path.

9.5 S1b) Equivalent impedance

The impedance $Z = U/I$ seen looking into the *passive* network from one bus, with the sources switched off. In S1, this is the factory seen from its feeding point.

Option	Kind	Meaning
Seen from bus `bus`	bus	Any bus of the network.
Cross-check against port `far_bus`	bus	Optional: recompute through the two-port relation and report the residual.

The impedance comes from Kron reduction of the Y-bus onto the bus. With a **Cross-check against port**, it is computed again through the two-port relation $Z = (AZ_{\text{term}} + B)/(CZ_{\text{term}} + D)$, with Z_{term} the termination at the far port, and both residuals are reported.

Result. Z in per unit and in ohm, $|Z|$, $\arg Z$ and the base impedance, then the cross-check table.

9.6 S1c) Bus voltages and the linear network

Two studies share one solve: **S1c) Bus voltages** and **Bus voltages (linear network)**. They solve the network driven by its sources, with every load a constant impedance, so one linear solve is exact:

$$YU = I, \qquad \text{ideal sources fixing their bus voltage,} \quad \text{others a Norton pair } \left(\tfrac{1}{Z_{\text{th}}},\ \tfrac{E}{Z_{\text{th}}}\right).$$

This is the pre-fault state the Thevenin and fault studies start from.

Option	Kind	Meaning
Report voltage at bus `bus`	bus	Any bus of the network.

Bus voltages (linear network) has the same option:

Option	Kind	Meaning
Report voltage at bus `bus`	bus	Any bus of the network.

Result.

- the headline: the voltage at the chosen bus in pu, angle and kV, and the voltage drop from the source in percent;
- **Phasor chain** (single-source networks): the S1 chain $U_a \to I_a \to [U_e; I_e]$, with $I_a = U_a/Z_I$ and $[U_e; I_e]$ from the two-port matrix;
- **Bus voltages**: every bus in pu, angle, kV, and the current injected;
- **Branch flows**: each branch's current, power and losses at its sending end, with the total losses in the note;
- **Sources**: each source's current in pu and kA, and its power in MVA;

- **Cross-check**: the far-bus voltage from the nodal solve against the two-port chain, and the source current against U/Z.

With more than one source, the phasor chain and the two-port cross-check are skipped, with a warning, because $I = U/Z$ then no longer holds. The nodal solve is exact either way.

Annotations. Every bus shows its voltage in kV and in pu with its angle, and every branch its current and sending-end power.

9.7 Thevenin equivalent

The equivalent seen from one bus: the open-circuit voltage and the impedance of each sequence network.

Option	Kind	Meaning
Seen from bus bus	bus	Any bus of the network.

How. U_{th} is the bus voltage of the pre-fault linear solve, with no voltage factor. $Z_{\text{th},1}$, $Z_{\text{th},2}$ and $Z_{\text{th},0}$ are the diagonal entries Z_{kk} of the impedance matrices of the three sequence networks, with the sources' EMFs switched off and their impedances kept. They are computed as columns of $Z = Y^{-1}$ from one LU factorisation per sequence. A bus with no path to earth in a sequence, such as the zero sequence behind a delta winding, has an *open* (infinite) impedance there. When a line has no zero-sequence data, $Z_{\text{th},0}$ is left out with a warning.

Cross-checks. $Z_{\text{th},1}$ is compared with the Kron reduction of the Y-bus onto the bus. With a single source behind an impedance, U_{th} is also compared with the route of the MATLAB solution, $U_{\text{th}} = Z(k,s)\,E/Z(s,s)$. The residuals are in the **Cross-checks** table.

Result. U_{th} in pu and kV; each Z_{th} in pu and ohm; the three-phase fault current $I_{sc} = U_{\text{th}}/Z_{\text{th},1}$ in pu and kA; the short-circuit power as the course defines it, $S_{sc} = U_{\text{th}}\,I_{sc}{}^{*}\,S_{\text{base}}$, its magnitude, and $\sqrt{3}\,U_{\text{base}}|I_{sc}|$ for comparison; and, when Z_0 is known, the line-to-ground fault current $3U_{\text{th}}/(Z_1 + Z_2 + Z_0)$.

Annotations. The bus shows U_{th}, $Z_{\text{th},1}$ and $Z_{\text{th},0}$.

Note

A bus that holds an ideal source has $Z_{\text{th}} = 0$ by construction. PowerLab refuses it and asks for a finite source impedance. The default bus avoids such a bus.

9.8 Short circuit

A shunt fault at one bus, with every bus voltage and branch current during the fault, phase by phase.

Option	Kind	Meaning
Faulted bus `bus`	bus	Any bus of the network.
Fault type `kind`	choice	3ph: three-phase. slg: phase a to ground. ll: phases b and c. llg: phases b and c to ground. One of `3ph`, `slg`, `ll`, `llg`.
Fault impedance Z_f [ohm] `zf_ohm`	complex	Per faulted phase; `0` is a solid fault.
Ground impedance Z_g [ohm] `zg_ohm`	complex	Common path to earth of a double line-to-ground fault. Shown when **Fault type** is `llg`.

Fault types. With phase a as the reference phase:

`3ph` all three phases to a common point through Z_f each;

`slg` phase a to earth through Z_f;

`ll` phases b and c together through Z_f;

`llg` phases b and c to earth, each through Z_f and together through Z_g.

How. Each fault is written as three phase equations $A\,U_{abc} + B\,I_{abc} = 0$ and solved in symmetrical components against the sequence Thevenin equivalents of the bus (Section A.5). For a solid single line-to-ground fault this reduces to the textbook $I_1 = I_2 = I_0 = U_{\text{th}}/(Z_1 + Z_2 + Z_0 + 3Z_f)$. Only the sequences a fault involves are built: a three-phase fault needs neither the negative nor the zero sequence. The voltages at every other bus follow from one impedance-matrix column per sequence, $U_n(i) = U_{\text{pre},n}(i) - Z_n(i,k)\,I_n$, and the branch currents from those voltages. There is no voltage factor: the pre-fault state is the linear network.

Result.

- **Fault**: bus, type, Z_f (and Z_g) in ohm and pu, U_{pre}, the sequence impedances, I_1, I_2, I_0 and I_f in pu and kA, the earth current $3I_0$ for faults to earth, the fault power $S_f = U_{\text{th}} {I_f}^* S_{\text{base}}$ for a three-phase fault, and $\sqrt{3}\,U_{\text{base}}|I_f|$;
- **At the faulted bus**: the phase currents and line-to-neutral voltages;
- **Bus voltages during the fault**: U_a, U_b, U_c of every bus in pu and kV;
- **Branch currents during the fault**: each branch's phase currents and its largest current in kA;
- the Thevenin cross-checks.

Annotations. The faulted bus shows I_f in kA and its three phase voltages. Other buses show their voltage, or their three phase voltages when the fault unbalances them, and every branch shows its largest current.

Tip

For a quick three-phase fault, right-click a bus and choose **Three-phase short circuit here**.

9.9 Load flow (Newton–Raphson)

The steady state of the network with its generators, voltage controllers and constant-power loads: the problem of Assignment LF.

Option	Kind	Meaning
Load model `load_model`	choice	as_set: each load's own model. constant_power: every load is a PQ injection. constant_impedance: every load is an admittance at its nominal voltage. One of `as_set`, `constant_power`, `constant_impedance`.
Shunt model `shunt_model`	choice	as_set: each shunt's own model. constant_q: the shunt injects its rated Q whatever the voltage. constant_b: the rating fixes B at U nominal, so it delivers B U^2. One of `as_set`, `constant_q`, `constant_b`.
Enforce reactive limits `q_limits`	check box	A PU bus whose generators or SVC would leave their Q limits becomes a PQ bus held at the limit.
Tolerance [pu] `tolerance`	number	Largest power mismatch accepted. At least 0.
Maximum iterations `max_iter`	integer	At least 1.

Bus types. PowerLab derives each bus's type from what is connected, and names it as the course does:

Slack a bus with a slack generator or an external grid with **Slack** checked. Its voltage and angle are given. Exactly one per island is required.

PU a bus with a PV generator or an SVC. Its active power and voltage magnitude are given.

PQ every other bus. Its active and reactive power are given (zero at a bus with nothing connected).

What enters where. Constant-impedance loads and constant-B shunts are part of the admittance matrix. Constant-power loads, constant-Q shunts, PQ generators and the fixed P of PV generators are *injections*, with generation counted positive. External grids enter only as slack buses; their impedance is short-circuit data.

The method. Polar Newton–Raphson from a flat start (1 pu and 0° except at the given voltages). The mismatches are

$$\Delta P_k = P_k^{\text{spec}} - P_k(\theta, U) \;\; \text{(PU and PQ buses)}, \qquad \Delta Q_k = Q_k^{\text{spec}} - Q_k(\theta, U) \;\; \text{(PQ buses)},$$

and each iteration solves for the angle and relative magnitude corrections

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} H & N \\ J & L \end{bmatrix} \begin{bmatrix} \Delta\theta \\ \Delta U/U \end{bmatrix}, \qquad \theta \mathrel{+}= \Delta\theta, \quad U \mathrel{*}= 1 + \Delta U/U,$$

with $H = \partial P/\partial\theta$, $N = U\,\partial P/\partial U$, $J = \partial Q/\partial\theta$ and $L = U\,\partial Q/\partial U$ (the element formulas are in Section A.6). The iteration stops when the largest mismatch falls below the tolerance (10^{-10} pu by default), usually in four or five iterations.

Reactive limits. With **Enforce reactive limits** checked, a PU bus whose generators or SVC would need more than their combined **Q max** (or less than **Q min**) becomes a PQ bus, held at that limit, and the flow is solved again from the current state. A warning names the bus. A bus that reaches a limit stays PQ: there is no switching back in this version.

Result.

- the headline: convergence, iterations, the total losses and the slack's P and Q;
- **Bus types and voltages**: every bus's type, $|U|$ in pu and kV, angle, and the net P and Q it injects;
- **Generation and voltage control**: each generator, external grid and SVC with its bus type, P and Q. An SVC also reports $B = Q/U^2$ in pu and siemens, and a limited device says which limit holds it;
- **Loads**: each load's model and the P and Q it draws at the solved voltage. A constant-impedance load draws $S_n\,U^2$;
- **Shunts**: each shunt's model, the Q it delivers and its B. A constant-B shunt delivers $B\,U^2$, which is why the LF variant's bank at bus 5 gives 8.3648 MVAr rather than its rated 8.2 MVAr;
- **Branch flows**: P and Q into each branch at both ends, its active and reactive losses, and its largest current in kA;
- **Losses and power balance**: P_{loss} as the sum of the branch losses, Q_{loss}, the generation and the load, and the two balance residuals, which close at the tolerance;
- **Newton-Raphson iterations**: for each iteration, the largest $|\Delta P|$ and $|\Delta Q|$, the bus where the larger one occurs, and the largest step taken;
- **Jacobian [H N; J L] at the flat start**, with every row and column labelled. The Jacobian at the solution and the load-flow Y-bus are in the result too, and they come out with **Export results** as JSON.

When it does not converge. The study does not stop with an error. It shows the last iterate with a warning, *"The load flow did not converge: largest mismatch … after 20 iterations. The values below are the last iterate, not a solution."*, and keeps the iteration log, so you can see where it went wrong. Typical causes are a load far beyond what the network can carry, or an island whose only voltage reference is missing.

Annotations. Every bus shows $|U|\angle\theta$, its type and its voltage in kV. Voltages outside 0.95–1.05 pu are drawn in the warning colour. Every branch shows its sending-end P and Q and its losses, every generator its P and Q, and every SVC its Q and B.

9.10 Unbalanced load

A load that draws a different power on each phase, solved with the sequence networks. Assignment S3e disconnects half of phase a of load LDe: factors 0.5, 1, 1.

Option	Kind	Meaning
Load `element`	a load	
Phase a factor `factor_a`	number	Share of the load's phase-a power still connected; 0.5 is half of it disconnected. At least 0.
Phase b factor `factor_b`	number	At least 0.
Phase c factor `factor_c`	number	At least 0.
Report voltages at bus `bus`	bus	Any bus of the network.

How. The load is taken out of the sequence networks, which then give the voltage at its bus without it and the sequence Thevenin impedances there. The load's phase admittance matrix Y_{ph}, built from its balanced admittance, its connection and the phase factors, closes the problem, solved like a fault with $A = Y_{\text{ph}}$ and $B = -\mathbf{I}$ (Section A.5). For a grounded star, Y_{ph} is diagonal; a neutral impedance couples the phases; an ungrounded star and a delta carry no zero-sequence current. For a delta, the factors apply to the branches ab, bc and ca. The voltages elsewhere follow from the impedance-matrix columns.

Result. The load (connection, factors, balanced impedance, voltage without it, sequence impedances at its bus); the matrices Z_{ph} and $Z_s = T^{-1} Z_{\text{ph}} T$; the sequence and phase currents it draws and the phase voltages at its bus; the voltages at the report bus (pre-disturbance, sequence and phase); and every bus's three phase voltages.

Annotations. Every bus shows its three phase voltages in pu. The load's bus and the report bus are highlighted.

Chapter 10

Reading results

10.1 The Results panel

Each case run gets a tab in the **Results** panel, titled with the case's title. The last tab, **Log**, holds the full text of every result, the validation messages and anything else worth keeping, all selectable and copyable. A result tab has three parts:

Headline the main answer in one or two lines, such as *U_th = 0.9348 pu < -3.0196 deg Z_th,1 = 0.1645 + 0.0677i pu.*

Warnings in yellow banners: anything you should know, for example an element left out, a zero sequence that could not be built, an override that names something gone, or a load flow that did not converge. A case that could not run shows only its reason.

Tables the numbers, each with its title and sometimes a note below it.

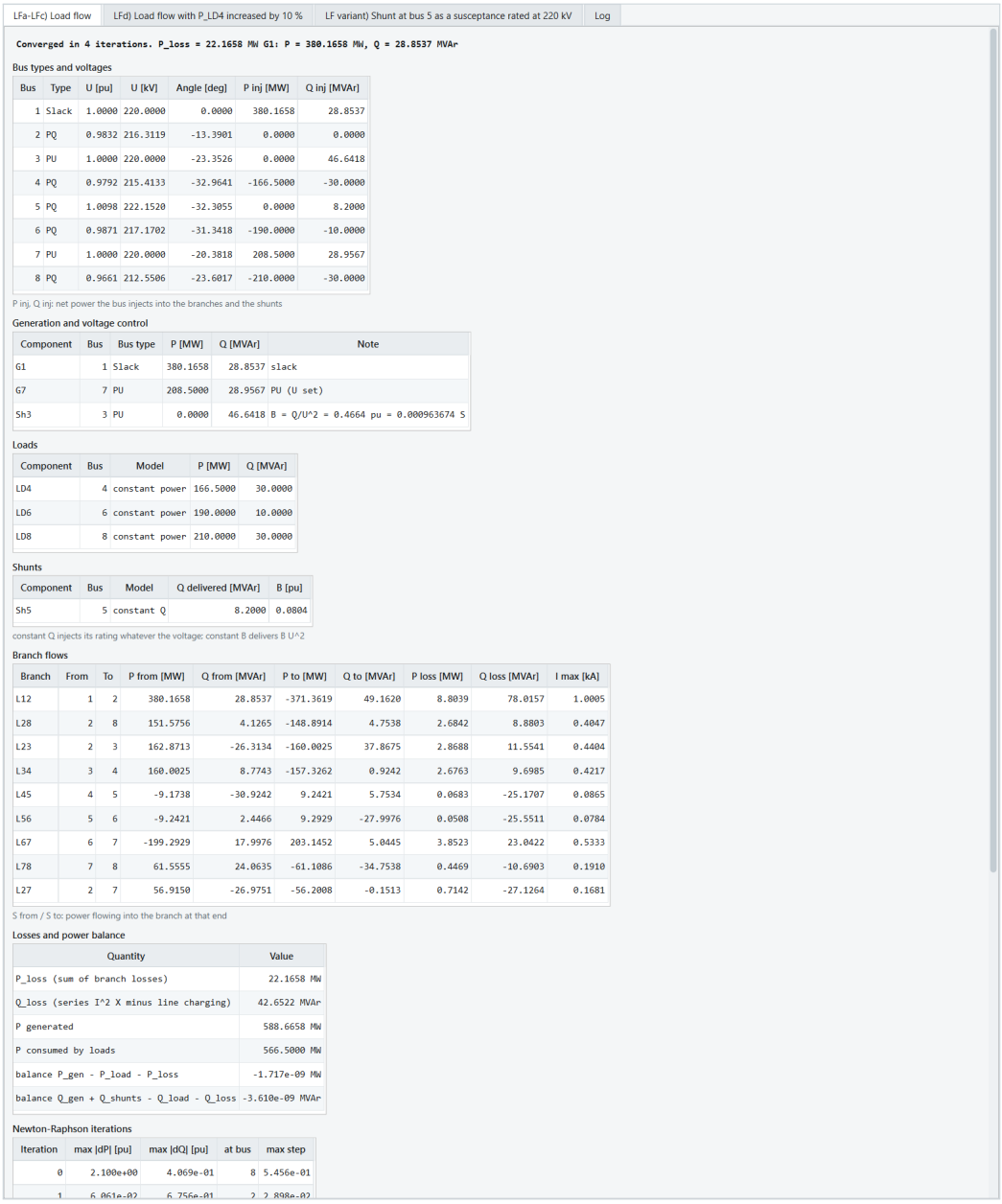


Figure 10.1: The first result tab of the load flow example (top part).

10.2 Number formats

PowerLab prints numbers the way MATLAB's `format short` does, so they can be compared with a MATLAB solution digit by digit and pasted into a report:

- four decimals in fixed point: `0.9832`, `22.1658`;
- scientific notation when a value is very large or very small: `3.6037e+03`, `1.2830e-05`;
- impedances, admittances and complex powers in rectangular form, `0.1645 + 0.0677i`; voltages and currents in polar form with the angle in degrees, `0.9952 pu < -0.5345 deg`;

- matrices with one common scale, as MATLAB prints them. When the largest entry leaves the fixed-point window, the title carries the factor (`1.0e+02 *`) and the entries are divided by it.

Units are always given: in the column header, after the value, or in the table's note.

10.3 Copying

- Select cells and press Ctrl + C to copy them as tab-separated text, ready for a spreadsheet.
- Right-click a table for **Copy**, the selection, or **Copy table**, all of it with its header row.
- The headline can be selected with the mouse and copied.

10.3.1 Copy as MATLAB

A table that prints a matrix (an admittance matrix, the load-flow Jacobian, the S1a matrix F, the Z_{ph} and Z_s of an unbalanced load) offers **Copy as MATLAB** in its context menu. It copies the matrix as MATLAB source with every digit kept, not the four decimals on screen. The two-port matrix of S1, for example, copies as:

```
F = [1.0002045432793645 + 0.0004317301765735776i, 0.004880874748045769 + 0.27917704044800307i;
     ...
     0.00369999999509357 - 0.0017535002966172987i, 1.0003032467164057 + 0.0005924141613021 07i];
```

Paste it into MATLAB and compare it with your own matrix to the last digit.

10.4 Annotating the diagram

Most studies can annotate the diagram with their result. After a run, the selected result tab's annotations are drawn next to the elements they describe:

- bus annotations above the bus, in green;
- branch annotations below the branch, beside it for a vertical branch, in brown;
- element annotations, such as a generator's P and Q, to the right of the symbol;
- values that call for attention, such as a faulted bus or a voltage outside 0.95–1.05 pu, in the warning colour.

Click another result tab and the diagram shows that result instead. View › Show results on the diagram (Ctrl + R) hides or shows the annotations. They follow their element when it moves, and they disappear when the model changes, because they no longer describe it. Each study's section in Chapter 9 says what it draws. The model, sequence model, admittance matrix, S1a and S1b results are not drawn on the diagram.

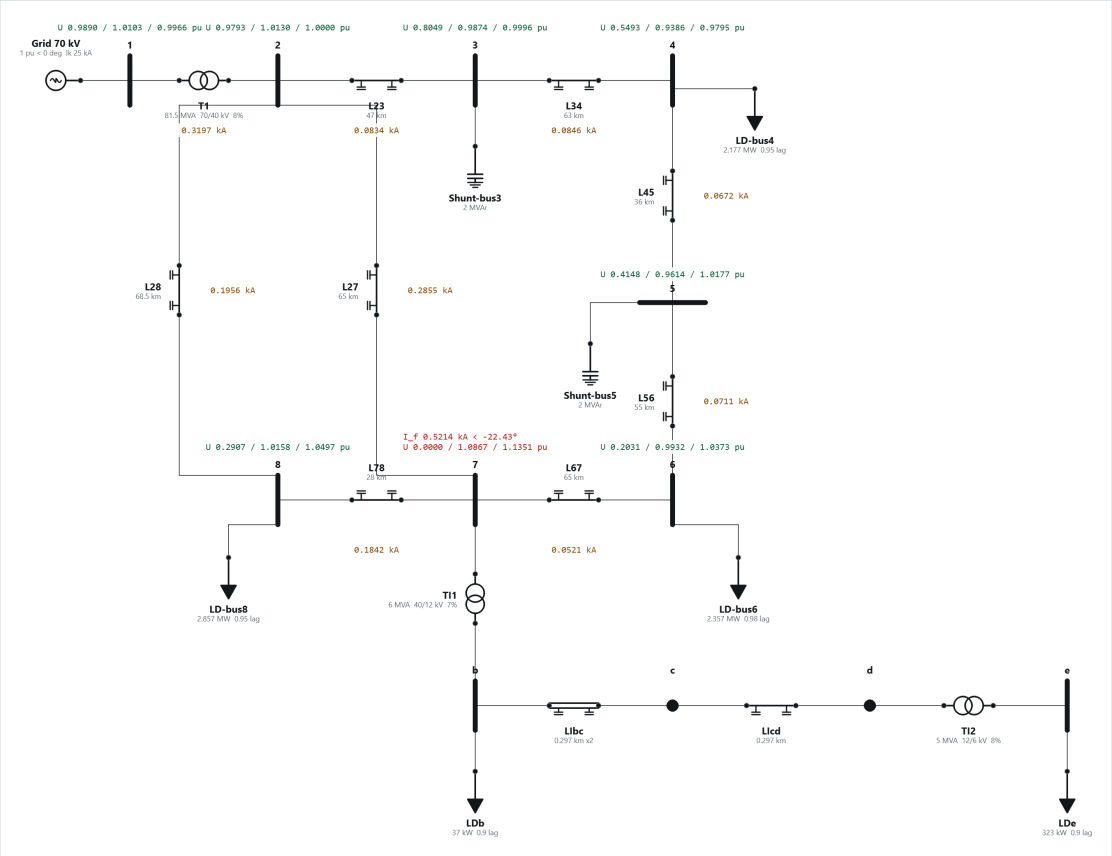


Figure 10.2: A single line-to-ground fault at bus 7 of `grid_industry.psn` (case S3c). The faulted bus shows the fault current and its three phase voltages, the other buses their phase voltages during the fault, and the branches their largest current.

10.5 Colouring the diagram

View › Colour by colours the diagram, with a legend in the top-right corner of the canvas:

Nothing every element in black.

Voltage level one colour per base voltage, highest first. Buses, the elements that sit in one voltage zone and the wires that land on each bus take their zone's colour. Transformers, which join two zones, stay black. The colours follow edits: a new element is coloured as soon as it is connected.

Voltage result every bus coloured by the band its voltage magnitude falls in, for the result tab shown: below 0.90, 0.90–0.95, 0.95–0.98, 0.98–1.02, 1.02–1.05 and above 1.05 pu. During a fault, the lowest phase counts. A result without bus voltages, such as an admittance matrix, leaves the buses black and says so in the status bar.

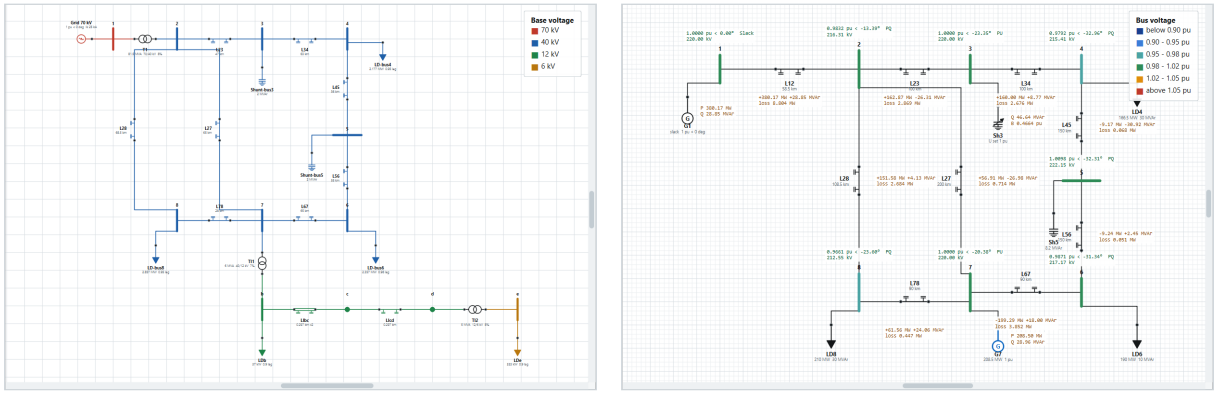


Figure 10.3: Left: the S2 grid with the S1 industry coloured by voltage level (70, 40, 12 and 6 kV). Right: the load flow coloured by bus voltage. Bus 8, at 0.9661 pu, falls in the 0.95–0.98 band.

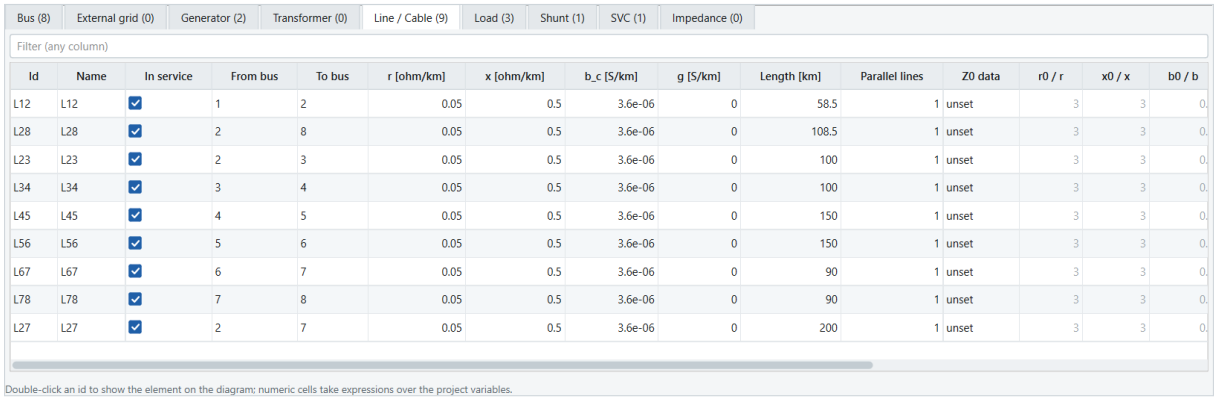
10.6 Validation messages

Before every run, PowerLab validates the network. Errors stop the run and are listed in a message box and in the **Log**. A component that is not wired does not stop the run: the studies leave it out, and every result says so in a warning, *"LD9 is not connected to a bus, so the studies ignore it."* Chapter 16 lists the messages with what to do.

Chapter 11

The data manager

The data manager shows every element of a type in one editable table, the way PowerFactory's data manager or a spreadsheet does. Use it to check the data of a whole network at a glance, or to change many elements without clicking each one on the canvas.



Id	Name	In service	From bus	To bus	r [ohm/km]	x [ohm/km]	b_c [S/km]	g [S/km]	Length [km]	Parallel lines	Z0 data	r0 / r	x0 / x	b0 / b
L12	L12	☑	1	2	0.05	0.5	3.6e-06	0	58.5	1	unset	3	3	0.
L28	L28	☑	2	8	0.05	0.5	3.6e-06	0	108.5	1	unset	3	3	0.
L23	L23	☑	2	3	0.05	0.5	3.6e-06	0	100	1	unset	3	3	0.
L34	L34	☑	3	4	0.05	0.5	3.6e-06	0	100	1	unset	3	3	0.
L45	L45	☑	4	5	0.05	0.5	3.6e-06	0	150	1	unset	3	3	0.
L56	L56	☑	5	6	0.05	0.5	3.6e-06	0	150	1	unset	3	3	0.
L67	L67	☑	6	7	0.05	0.5	3.6e-06	0	90	1	unset	3	3	0.
L78	L78	☑	7	8	0.05	0.5	3.6e-06	0	90	1	unset	3	3	0.
L27	L27	☑	2	7	0.05	0.5	3.6e-06	0	200	1	unset	3	3	0.

Figure 11.1: The lines of the load flow example in the data manager. The **r0 / r** column and the ones after it are greyed out: those fields only apply when **Z0 data** is `factors`.

11.1 Opening it

The data manager shares the bottom panel with **Results**: click its tab, or use View › Data manager (Ctrl + ⇧ + D). The menu entry brings the panel forward on the type of the selected element.

11.2 The tables

There is one tab per component type, in library order, titled with the number of elements of that type: **Bus (8)**, **Line / Cable (9)**, … Each row is an element, in network order. The first column is its id, and the others are its fields, with the same labels and units as the property editor. Hover a column header to read the field's description.

- **Greyed-out cells** belong to fields the element's mode does not use, such as a load's **cos phi** when it is given by P and Q, or a line's zero-sequence factors when its **Z0 data** is `absolute`. They are still editable.

- **Tinted cells** hold a value that comes from an expression (a binding). Hover the cell to see the expression.
- **Bus columns** show the bus's name.

11.3 Editing

Double-click a cell, or select it and start typing. Cells take the same input as the property editor:

Numbers and expressions A number is stored as it is. An expression over the project variables becomes a binding. An empty cell removes the binding, and makes the value empty for fields that accept that, such as **U nominal** or **Q max**.

Choices are picked from a list.

Check boxes (**In service**, **Slack**, …) toggle with a click.

Buses are picked from the list of buses, or typed by name or id. Pick **(none)** to disconnect the terminal.

Every edit is one undoable step. The canvas and the property editor follow it at once, and the results become stale. An edit that is not valid, such as text in a numeric field, a negative length or a bus that does not exist, is refused. The reason appears in the status bar.

11.4 Sorting and filtering

- Click a column header to sort by that column; click it again to reverse the order.
- Type in the **Filter** box to keep the rows where any cell contains the text, case aside: `L2` keeps L23, L27 and L28.

Sorting and filtering only change the view, never the network.

11.5 Finding an element on the diagram

Double-click an element's id to select it on the canvas, centre the diagram on it and show its properties. Edit › Find element… (Ctrl + F) does the same from anywhere (Section 4.11).

Chapter 12

Exporting

12.1 Results

File › Export results... (Ctrl + E) writes the results of the last run, every result tab, to one file. The format follows the file type chosen in the dialog, or the extension you type:

Format	Extension	Contents
Text	`.txt`	Every result as on screen: title, headline, tables and warnings, in `format short`. The same text as the **Log**.
CSV	`.csv`	Every table as a block: its title, the header row, the rows and its note, then a blank line. Opens in any spreadsheet.
LaTeX	`.tex`	A `\subsection*` per result and a `booktabs` tabular per table, with LaTeX's special characters escaped. The polar < becomes ∠, and degrees are typeset properly. Needs `\usepackage{booktabs}` in the report's preamble.
JSON	`.json`	The numbers behind the tables at full precision: for each result, its scalars, per-bus and per-element quantities (value, kind and label), every matrix with its row and column labels, and the tables. Complex numbers are written as `{"re": ..., "im": ...}`.

Tip

Use LaTeX for a report, and JSON when a script of your own should read the numbers, including the matrices that are not printed as tables, such as the Jacobian at the load-flow solution. Text and CSV carry the four decimals shown on screen; JSON carries every digit.

If nothing has been run yet, PowerLab says so in the status bar instead of writing an empty file.

12.2 The diagram

File › Export diagram... (Ctrl + ⇧ + E) saves the drawing:

PNG a bitmap at twice the screen scale, sharp enough to print at a column's width;

SVG a vector drawing, for editing or for the web;

PDF a vector page exactly the size of the drawing, for `\includegraphics` in LaTeX.

The export contains every element plus a margin, on white, without the grid and without the selection highlight. The annotations of the result shown are included when they are on (Ctrl + R), which makes the export a ready figure for a report. The figures of this manual that show diagrams were exported this way.

12.3 Matrices for MATLAB

Copy as MATLAB, in the context menu of a matrix table, puts the matrix on the clipboard as MATLAB source with every digit (Section 10.3.1). The JSON export holds every matrix of every result.

12.4 From the command line

`run_headless.py --export results.json` writes the same files without opening the window (Chapter 13).

Chapter 13

Working from the command line

Two scripts in the repository's root solve projects without opening a window. They do not need PySide6.

13.1 run_headless.py

`run_headless.py` runs the study cases of any project, prints every result and can export what it ran:

```
python run_headless.py examples/lf_220kv.psn                  # every case of the project
python run_headless.py examples/lf_220kv.psn --case LFd       # one case (repeatable)
python run_headless.py examples/s2_grid.psn --study thevenin  # a study with its defaults
python run_headless.py examples/lf_220kv.psn --set A=0        # change a variable first
python run_headless.py examples/s2_grid.psn --export s2.json  # also write a file
python run_headless.py examples/s2_grid.psn --list            # cases and studies
```

Option	What it does
`project.psn`	The project to solve.
`-case ID`	Run this case of the project. Repeatable. Without `-case` or `-study`, every case runs, or every study with its defaults if the project has no cases.
`-study KEY`	Run this study: the project's case of the same id if there is one, otherwise the study with its defaults. Repeatable. The keys are `model`, `s1a`, `s1b`, `s1c`, `ybus`, `linear`, `thevenin`, `short_circuit`, `loadflow`, `seqmodel` and `unbalanced_load`.
`-set NAME=VALUE`	Set a project variable before solving. Repeatable. `-set A=0` gives the answers printed in the assignments.
`-export PATH`	Also write the results to a file. The extension (`.txt`, `.csv`, `.tex`, `.json`) picks the format (Section 12.1).
`-quiet`	Do not print the results; useful with `-export`.
`-list`	List the project's cases (id, study, title) and the available studies, and stop.
`-run-script`	Run the variable script the project links to. Without it, the values saved in the project are used.

The output starts with the project's title, then the variables set and a note if a linked script was not run, then each result as in the **Log** tab.

Exit codes. `0` when every case ran. `1` when at least one case failed: its reason is printed and the other cases still run. `2` when the project cannot be opened, a `–case` names a case that does not exist, a `–set` is malformed or its expression fails, or the export file has an unknown extension. This makes the runner usable in scripts and batch checks:

```
python run_headless.py examples/lf_220kv.psn --case LFa --set A=0 --quiet --export lf_a0.json
```

13.2 run_s1_headless.py

`run_s1_headless.py` prints the S1 report: bases, per-unit values, S1a, S1b and S1c. It runs on the built-in S1 network or on a project.

```
python run_s1_headless.py                           # built-in S1 network, A = 80
python run_s1_headless.py --a-number 0              # the answer published in the statement
python run_s1_headless.py examples/s1_a80.psn       # a saved project
python run_s1_headless.py my_network.psn --study s1b  # a single study (repeatable)
```

Option	What it does
`project.psn`	The project; without it, the built-in S1 network.
`–a-number N`	The A-number of the built-in network (80 by default). Ignored for a project.
`–study KEY`	Only this study, any key. Repeatable. By default `model`, `s1a`, `s1b` and `s1c`.
`–run-script`	Run the project's linked variable script.

Each study uses the options of the project's case of the same id, so the ports and buses chosen in the window apply. It prints the network's validation messages first and returns exit code 2 when the project cannot be opened or the studies cannot run.

13.3 From Python

The engine is a plain Python package (`psa.core`) with no Qt in it, so a script can use it directly:

```
from psa.core.serialize import load
from psa.core.study import run_case
from psa.core.export import export_results

proj = load("examples/lf_220kv.psn")
proj.params.set_var("A", "0")
proj.apply_parameters()
results = [run_case(proj.network, case, proj.params) for case in proj.cases]
print(results[0].scalars["P_loss_MW"].value)       # 26.424...
export_results(results, "lf.json")
```

A `Result` holds the formatted tables and the numbers behind them: `scalars`, per-bus `buses`, per-element `elements` and labelled `matrices`. These are the same names as in the JSON export.

Chapter 14

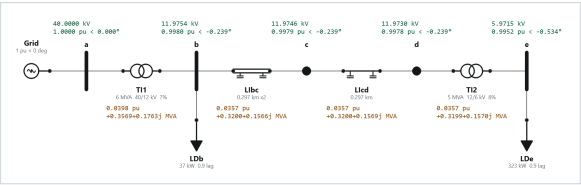
Solving the EG2100 assignments

Each assignment of the course is an example project whose study cases are its tasks. Open the project, press F5, and read each task in its result tab. Ctrl + R shows the selected result on the diagram. The console gives the same results: `python run_headless.py examples/<file>.psn`.

Every project is parametric in the A-number, the variable `A`. The values below are for $A = 80$, checked digit by digit against the MATLAB reference solutions. Set `A = 0` in the **Variables** panel (or pass `-set A=0`) to reproduce the check values printed in the assignment statements. The test suite checks those too.

14.1 S1: the factory network

`examples/s1_a80.psn`. A 40 kV grid (bus a) feeds a factory through transformer TI1 (6 MVA, 40/12 kV), two cable sections to bus d, and transformer TI2 (5 MVA, 12/6 kV) to bus e. Loads LDb and LDe depend on `A`, and so do the cable lengths.



Task	Case (study)	Result at $A = 80$
Per-unit model	`model`	U_{base} = 40, 12, 6 kV; $Z_{\text{TI1}} = j0.1167$ pu, $Z_{\text{TI2}} = j0.1600$ pu
S1a: two-port matrix from bus a to bus e	`s1a` (two-port)	$A = 1.0002 + j0.0004$, $B = 0.0049 + j0.2792$, $C = 0.0037 - j0.0018$, $D = 1.0003 + j0.0006$, $\det F = 1$
S1b: impedance seen from bus a	`s1b` (equivalent impedance)	$Z_I = 22.5232 + j11.1298$ pu ($A = 0$: $22.5156 + j11.0767$ pu)
S1c: voltage at bus e	`s1c` (bus voltages)	$U_e = 0.9952\angle -0.5345°$ pu $= 5.9715$ kV

14.2 S2: the meshed 40 kV grid

`examples/s2_grid.psn`, or `examples/grid_industry.psn`, where the S1 industry hangs from bus 7. In the latter, cases S2a–S2c take TI1 out of service (their **Network variant** tab), as the assignment does. The 70 kV grid is an external grid of 25 kA at $\cos\varphi = 0.15$.

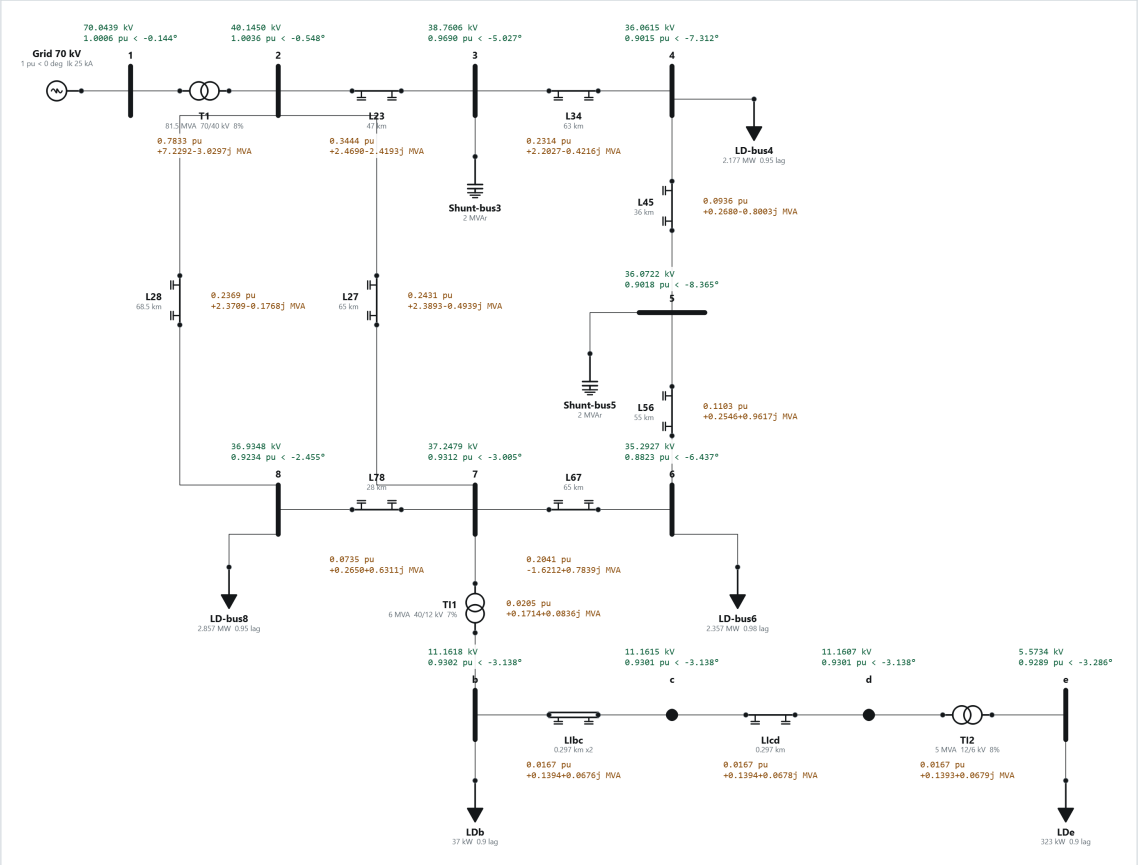


Figure 14.1: `grid_industry.psn` after case S2d: the industry's voltages with LDe at 50 %.

Task	Case (study)	Result at $A = 80$
S2a: admittance matrix with the 70 kV Thevenin source	`S2a` (Y-bus, source as internal node)	The 9×9 matrix with MATLAB's common factor; $Y(1,1) = 45.4663 - j401.5545$, $Y(9,9) = 45.4663 - j299.6795$ pu
S2b: Thevenin equivalent at bus 7	`S2b` (Thevenin)	$U_{\text{Th7}} = 0.9348\angle -3.0196°$, $Z_{\text{Th7}} = 0.1645 + j0.0677$ pu ($A = 0$: $0.9732\angle -1.4224°$, $0.0649 + j0.0356$ pu)
S2c: three-phase fault at bus 7	`S2c` (short circuit, `3ph`)	$I_{sc} = 0.7584$ kA, $S_{sc} = 45.4220 + j18.6951$ MVA
S2d: industry voltages with LDe at 50 %	`S2d` in `grid_industry.psn` (bus voltages, override `LDe.p_mw = 0.5*value`)	$U_{\text{LDe}} = 0.9289\angle -3.2857°$ pu $= 5.5734$ kV

14.3 S3: sequence networks and unsymmetrical conditions

`examples/grid_industry.psn`. The zero-sequence data is in the **Zero sequence** group of each element's properties: grid lines with $r_0 = 2.5r$, $x_0 = 2x$, $b_0 = 0.5b$; cables with $Z_0 = 3Z_1$, $b_0 = 0.5b$; TI2 a `Dyn` transformer; the bus-6 load and the bus-3 shunt delta-connected; the grid's Z_0 from $I_{k1} = 13$ kA at $\cos\varphi = 0.15$. S3a–S3c are also in `s2_grid.psn`.

Task	Case (study)	Result at $A = 80$
S3a: zero-sequence admittance matrix	`S3a` (Y-bus, `zero`, sources `grounded`)	The 8×8 matrix; $Y_0(7,7) = 4.8795 - j1.4449$ pu
S3b: sequence Thevenin impedances at bus 7	`S3b` (Thevenin)	$Z_{\text{Th7,1}} = Z_{\text{Th7,2}} = 0.1645 + j0.0677$, $Z_{\text{Th7,0}} = 0.4031 + j0.1226$ pu
S3c: single line-to-ground fault at bus 7	`S3c` (short circuit, `slg`)	$I_f = 3.6126\angle -22.4347°$ pu $= 0.5214$ kA
S3d: Thevenin equivalents at bus e, LDe disconnected	`S3d` (Thevenin, LDe out of service)	$U_{\text{Th,e}} = 0.9340\angle -3.0422°$, $Z_1 = 0.1694 + j0.3466$, $Z_0 = j0.1600$ pu
S3e: half of phase a of LDe disconnected	`S3e` (unbalanced load, factors 0.5, 1, 1)	Bus 7: $21.4825\angle -3.0011°$, $21.4474\angle -123.0463°$, $21.4503\angle 117.0576°$ kV

Note

The check values printed in the statement for S3b at $A = 0$ contain two misprints. It gives $Z_{\text{Th7,1}} = Z_{\text{Th7,2}} = 0.0694 + j0.0356$, where S2b prints $0.0649 + j0.0356$ for the same quantity. And it gives $Z_{\text{Th7,0}} = 0.1617 - j0.0656$, where the zero-sequence network seen from bus 7 is inductive: $0.1617 + j0.0656$. PowerLab matches the MATLAB solution. The file `tools/refs/README.md` in the repository documents both.

14.4 LF: the Newton–Raphson load flow

`examples/lf_220kv.psn`: eight 220 kV buses on 100 MVA. G1 at bus 1 is the slack, an SVC at bus 3 holds 220 kV (a PU bus with $P = 0$), and G7 at bus 7 delivers 208.5 MW at 220 kV. The loads at buses 4, 6 and 8 are constant power, and the bus-5 shunt injects its rated Q (the reading of the statement). The diagram on the title page of this manual is this project after case LFa.

Task	Case (study)	Result at $A = 80$
LFa: bus types and voltages	`LFa` (load flow)	Types Slack, PQ, PU, PQ, PQ, PQ, PU, PQ; for example $U_2 = 0.9832\angle -13.3901°$ pu
LFb: slack generation and the PU buses' reactive power	`LFa`	$P_{G1} = 380.1658$ MW, $Q_{G1} = 28.8537$, $Q_{sh,3} = 46.6418$, $Q_{G7} = 28.9567$ MVAr
LFc: total losses	`LFa`	$P_{\text{loss}} = 22.1658$ MW ($A = 0$: 26.4240 MW)
LFd: P_{LD4} up 10 %	`LFd` (override `LD4.p_mw = 1.1*value`)	$P_{\text{loss}} = 24.3993$ MW, $P_{G1} = 399.0493$ MW
The bus-5 shunt as a susceptance rated at 220 kV	`LFb_rated` (shunt model `constant_b`)	$P_{\text{loss}} = 22.1662$ MW, $Q_{sh,3} = 46.5526$ MVAr, $B_{sh,3} = 0.4655$ pu; the bank delivers 8.3648 MVAr at 1.0100 pu

The iteration log and the Jacobian at the flat start are at the bottom of each load-flow tab. The Jacobian can be copied as MATLAB.

14.5 From the assignments to your own networks

The projects are also templates. Copy one, change its data, and add cases. Drawing the S1 network from scratch is the tutorial of Section 4.14. The same steps apply to any radial or meshed network: place buses and elements, set the bases, add the variables your data needs, create a case per question, and run.

Chapter 15

Project files

A project is one `.psn` file: UTF-8 JSON with line feeds, readable in any text editor and easy to compare in a version-control diff. It holds the network, its variables, the drawing and the study cases. Results are not saved; run the cases again after opening a project.

15.1 Structure

Key	Contents
`format`, `version`	`"psa-project"` and the format version, 2 today.
`meta`	Title and notes of the project.
`network`	`s_base_mva`, `f_hz`, the base reference (`base_ref`: `bus` and `u_base_kv`), the `buses` and the `components`. Each component is an object with its `type`, its `id` and every field by its name. Complex numbers are written `[re, im]`.
`parameters`	`script_path`, the `project_vars` as `[name, expression]` pairs in order, and the `bindings` (`"LD4.p_mw": "158.5 + 0.1*A"`).
`layout`	The `grid` step, the `items` (each element's position, rotation, bus orientation and length, and a dragged label offset) and the `wires` (the bends of each wire, keyed `"COMPONENT:terminal"`).
`cases`	The study cases: `id`, `study`, `title`, `options`, `overrides` and `exclude`. Complex options are written `{"re": ..., "im": ...}`.
`view`	The display state, such as the annotations toggle.

A case as it appears in `examples/lf_220kv.psn`:

```
{"id": "LFd", "study": "loadflow",
 "title": "LFd) Load flow with P_LD4 increased by 10 %",
 "options": {}, "overrides": {"LD4.p_mw": "1.1*value"}}
```

15.2 Compatibility

- **Older files** are upgraded when opened. Version 1 kept per-study options instead of cases, and those become cases. The file is written in the current format the next time you save it.
- **New fields** have defaults, so a file written before a field existed opens with the default. A field the program does not know is ignored.

- **Only the network is required.** A file without a `layout` opens with an automatic layout: buses from left to right by distance from the base reference bus, branches between their buses and shunt elements below theirs.

15.3 The variable script

When a project links a variable script, the script stays a separate `.py` file. The project keeps its path and, under `project_vars` and in the fields, the values it gave last time. Opening the project never runs the script without asking (Section 7.5).

15.4 Saving and examples

Ctrl + S saves to the project's file. An untitled project, or one opened as a copy such as `run_gui.py -example`, asks for a name instead, so an example is never overwritten by accident. The window title shows the file's name, with an asterisk while there are unsaved changes. Closing the window, opening another project or starting a new one asks first whether to save them.

Chapter 16

Troubleshooting

PowerLab explains a problem in a message rather than with a traceback, and names the elements involved. This chapter lists the messages you are most likely to meet, grouped by when they appear, with what to do about each one.

16.1 Validation: before any study runs

These are checked on every run. An *error* stops the run, and a *warning* lets it go on.

Message	What to do
No base reference bus is set.	Network › Base settings...: pick the bus where you know the base voltage, and give that voltage.
The base voltage at the reference bus must be positive.	The same dialog: set **Base voltage there**.
X is not connected to a bus / at both ends, so the studies ignore it. (warning)	Wire the terminal (Section 4.2), or delete the element. Every result repeats this warning while it lasts.
X connects bus 'a' to itself.	Reconnect one end of the branch to another bus.
X: rated voltages must be positive. / rated power must be positive.	Fill in the transformer's ratings.
X has zero impedance. (warning; an error when a study converts it)	Give the transformer, line or impedance a non-zero r or x. A zero impedance is a short circuit between two buses: merge them instead.
X: cos phi must be in (0, 1].	Correct the load's power factor, or give it as P and Q.
X: the voltage setpoint must be positive. / Q min is above Q max.	Correct the generator's or SVC's setpoint or limits.
The network has N separate islands. (warning)	Usually intended, for example a case that disconnects part of the network. The network studies keep the islands that contain a source.

16.2 Bases

Message	What to do
Bus 'x' already has U_base = 12 kV but is also reached with 12.6 kV through 'T3'. Check the transformer ratios around this loop.	The rated ratios around a loop do not multiply to one. Correct a ratio, or pin the base you want on one bus (Chapter 6).

Bus 'x' cannot be reached from the base reference bus, so it has no per-unit base.	Connect the bus to the rest of the network, pin its base, or take it out of service.
X spans a change of voltage base (40 kV vs 12 kV). A line cannot change the base; use a transformer.	A line joins two zones. Use a transformer between them, or correct the bases.

16.3 Studies

Message	What to do
There is no source in the network, so the bus voltages are not defined. Add an external grid.	The linear, Thevenin, fault and unbalanced-load studies need an external grid. A generator only acts in the load flow.
X has no zero-sequence data. Set 'Z0 data' in its Zero sequence properties.	Give the line its zero-sequence data, as factors or per-km values. The data manager's **Line / Cable** tab sets it for many lines at once.
Bus 'x' holds an ideal source, so its impedance to ground is zero by construction. Give the source a finite impedance.	Give the external grid an impedance (**Z Thevenin**, or a fault level), or study another bus.
Bus 'x' is not energized in this case.	The case takes out of service the path from the sources to that bus. Check its **Network variant**.
The condition at bus 'x' cannot be solved: it short-circuits an ideal source or leaves the currents undetermined.	A fault at, or next to, an ideal source. Give the source an impedance.
The two ports must be different buses. (S1a)	Choose two different port buses.
Nothing is connected to bus 'x', so its equivalent impedance is infinite. (S1b)	Choose a bus with something connected, or connect it.
The network cannot be solved: some buses are floating (no path to ground and no source).	A part of the network has no shunt and no source: add the missing connection, or take the floating part out of service.

16.4 Load flow

Message	What to do
The island with buses … has no slack bus: connect an external grid or set a generator's control to slack.	Every island needs one slack bus.
The island with buses … has 2 slack buses; a load flow needs exactly one per island.	Set all but one to pv, or clear **Slack** on extra external grids.
Conflicting voltage setpoints at bus x: G7 and G7b ask for different values.	Give every voltage-controlling device on a bus the same **U set**.
The load flow did not converge. (warning)	Read the iteration log: a mismatch that grows points at the bus named in the **at bus** column. Typical causes are a load beyond what the network can carry, a voltage setpoint far from 1 pu, or wrong units, such as kW typed as MW. Raise **Maximum iterations** only when the mismatch was still falling.
Bus x reached its max reactive limit and becomes a PQ bus. (warning, with **Enforce reactive limits**)	Information: the bus can no longer hold its voltage. Its voltage is now a result.

16.5 Study cases

Message	What to do
Override 'x': write it as COMPONENT_ID.field, e.g. LD4.p_mw.	The dialog refuses an override whose element or field does not exist. Use the element's id, not its name, and a field name from Chapter 5.
Case 'c': override 'x' names something that does not exist, so it is ignored. (warning)	An element the case refers to was deleted or renamed. Edit the case.
Unknown name 'B'. Define it as a variable or load it from your script.	An expression uses a variable that does not exist. Add it in the **Variables** panel.

16.6 Installation and display

Qt fails to load with *"the specified procedure could not be found"*. On Windows with Anaconda, Anaconda's ICU library shadows the one Qt expects. Start PowerLab with `run_gui.py`, which imports `psa.gui` first; in your own scripts, import `psa.gui` before PySide6.

The window layout looks wrong Panels can be dragged back into place, or shown again from **View**. To return to the default layout, close PowerLab and delete the `windowState` value under `HKCU\Software\PowerLab\PowerLab` in the registry.

The results do not change after an edit Results are not recomputed by themselves. When the tabs carry an asterisk, press F5.

A pasted element is unconnected Its bus was not copied with it (Section 4.9). Wire it, or copy the bus too.

16.7 Limitations of version 1.0

- One diagram page per project. No breakers or switches as components (take elements out of service instead). No three-winding transformers.
- Transformer taps are stored but not modelled (an off-nominal tap only gives a warning), and there are no phase shifters. Phase shifts of vector groups are not modelled in the sequence networks.
- Short circuits are computed from the linear pre-fault state, as in the course, without the voltage factors of IEC 60909.
- The load flow has one slack per island (no distributed slack), and a PU bus that reaches a reactive limit stays PQ (no switching back).
- The data manager edits one cell at a time.

Appendix A

Modelling reference

This appendix collects the equations behind the studies, in the notation of the course. Everything is per unit on the bases of Chapter 6 unless a unit is given.

A.1 Components in per unit

Component	Model
Transformer	series $z = \dfrac{r+jx}{100}\,\dfrac{S_{\text{base}}}{S_n}\left(\dfrac{U_n}{U_{\text{base}}}\right)^2$ on the side chosen in **Refer Z to**; no magnetising branch; ratio absorbed by the bases.
Line (n in parallel)	PI: $z = \dfrac{(r+jx)\,\ell}{n\,Z_{\text{base}}}$, $\dfrac{y_{\text{sh}}}{2} = \dfrac{n\,(g+jb_c)\,\ell}{2}\,Z_{\text{base}}$ at each end.
Load (constant Z)	$y = \dfrac{S^*}{S_{\text{base}}}\left(\dfrac{U_{\text{base}}}{U_n}\right)^2$, $S = P + jQ$, $Q = P\tan\varphi$ (lag) or $-P\tan\varphi$ (lead).
Shunt (constant B)	$y = j\,\dfrac{Q}{S_{\text{base}}}\left(\dfrac{U_{\text{base}}}{U_n}\right)^2$, capacitive for $Q > 0$.
External grid	$E = U\angle\theta$ behind Z_{th}, with $\lvert Z_{\text{th}}\rvert = \dfrac{U}{\sqrt{3}\,I_k}$ or $\dfrac{U^2}{S_k}$ and $\arg Z_{\text{th}} = \arccos(\cos\varphi)$; $Z_2 = Z_1$; $Z_0 = \dfrac{3U}{\sqrt{3}\,I_{k1}}\angle\arccos(\cos\varphi_0) - 2Z_1$ (mode `ik1`).
Impedance	$z = (R+jX)/Z_{\text{base}}$ of the from-bus, or $R+jX$ in pu; the same in every sequence.

A.2 Zero-sequence paths

Element	Zero-sequence model
Line	PI from the **Z0 data**: factors of r, x, g, b_c, or per-km values.
Transformer YNyn	series $Z_0 + 3Z_{n,\text{HV}} + 3Z_{n,\text{LV}}$
Transformer YNd	$Z_0 + 3Z_{n,\text{HV}}$ from the HV bus to earth; LV side open
Transformer Dyn	$Z_0 + 3Z_{n,\text{LV}}$ from the LV bus to earth; HV side open
Other vector groups	open
Load or shunt, Y0	$y_0 = \dfrac{1}{1/y + 3Z_n}$
Load or shunt, Y or D	open

External grid	behind Z_0, no source voltage; open when **Z0 from** is none

Z_0 of a transformer comes from **r0** and **x0**, which default to r and x. The negative sequence equals the positive one with the sources' voltages set to zero. Phase shifts are not modelled.

A.3 The admittance matrix

Every branch contributes a 2×2 admittance block between its buses i and j. For a PI section with series z and half-shunt y_{sh}:

$$\begin{bmatrix} I_i \\ I_j \end{bmatrix} = \begin{bmatrix} \frac{1}{z} + y_{\text{sh}} & -\frac{1}{z} \\ -\frac{1}{z} & \frac{1}{z} + y_{\text{sh}} \end{bmatrix} \begin{bmatrix} U_i \\ U_j \end{bmatrix}.$$

Shunt admittances add to the diagonal. A source behind Z_{th} adds $1/Z_{\text{th}}$ to its bus's diagonal (`grounded`), or a series branch Z_{th} to an internal node (`internal_node`). An ideal source is never an admittance; it fixes its bus voltage.

A.4 Linear solve, impedance matrix and Thevenin

With the buses split into those fixed by ideal sources (f) and the rest (r), and the Norton currents $I = E/Z_{\text{th}}$ of the other sources,

$$Y_{rr}\, U_r = I_r - Y_{rf}\, U_f.$$

The impedance matrix of a sequence network is $Z = Y^{-1}$ with the ideal-source buses grounded. PowerLab computes the columns it needs from one LU factorisation. A bus with no path to earth in a sequence has an infinite (open) impedance there. The Thevenin equivalent at bus k is

$$U_{\text{th}} = U_k^{\text{pre}}, \qquad Z_{\text{th},s} = Z_s(k,k), \quad s = 1, 2, 0,$$

and the short-circuit power is $S_{sc} = U_{\text{th}}\, {I_{sc}}^*\, S_{\text{base}}$ with $I_{sc} = U_{\text{th}}/Z_{\text{th},1}$.

A.5 Faults and unbalanced loads

A shunt condition at bus k is three phase equations

$$A\, U_{abc} + B\, I_{abc} = 0,$$

with I_{abc} the currents drawn from the network. In the course's sequence order $(1, 2, 0)$,

$$U_{abc} = T\, U_s, \quad I_{abc} = T\, I_s, \qquad T = \begin{bmatrix} 1 & 1 & 1 \\ a^2 & a & 1 \\ a & a^2 & 1 \end{bmatrix}, \quad a = e^{j120^\circ},$$

and each sequence network seen from k is its Thevenin equivalent, $U_s = U_s^{\text{pre}} - Z_s I_s$ with $U_s^{\text{pre}} = (U_{\text{th}}, 0, 0)$. Substituting gives one 3×3 system,

$$(B\, T - A\, T\, Z_s)\, I_s = -A\, T\, U_s^{\text{pre}}.$$

A sequence in which the bus has no path to earth carries no current; its voltage becomes the unknown instead. The faults, with phase a as the reference:

Fault	Conditions	A	B
Three-phase	$U_p = Z_f I_p,\ p = a,b,c$	$\mathbf{I}$	$-Z_f\,\mathbf{I}$
Line to ground	$U_a = Z_f I_a,\ I_b = I_c = 0$	$\left[\begin{smallmatrix}1&0&0\\0&0&0\\0&0&0\end{smallmatrix}\right]$	$\left[\begin{smallmatrix}-Z_f&0&0\\0&1&0\\0&0&1\end{smallmatrix}\right]$
Line to line	$I_a = 0,\ I_b = -I_c,\ U_b - U_c = Z_f I_b$	$\left[\begin{smallmatrix}0&0&0\\0&0&0\\0&1&-1\end{smallmatrix}\right]$	$\left[\begin{smallmatrix}1&0&0\\0&1&1\\0&-Z_f&0\end{smallmatrix}\right]$
Double line to ground	$I_a = 0,\ U_{b,c} = Z_f I_{b,c} + Z_g(I_b + I_c)$	$\left[\begin{smallmatrix}0&0&0\\0&1&0\\0&0&1\end{smallmatrix}\right]$	$\left[\begin{smallmatrix}1&0&0\\0&-Z_f-Z_g&-Z_g\\0&-Z_g&-Z_f-Z_g\end{smallmatrix}\right]$
Unbalanced load	$I_{abc} = Y_{\text{ph}}\,U_{abc}$	Y_{ph}	$-\mathbf{I}$

Here $\mathbf{I}$ is the 3×3 identity matrix.

For a solid line-to-ground fault this reduces to $I_1 = I_2 = I_0 = U_{\text{th}}/(Z_1 + Z_2 + Z_0 + 3Z_f)$. The voltages at the other buses follow from $U_s(i) = U_s^{\text{pre}}(i) - Z_s(i,k)\,I_s$.

Phase admittance of a load. With y the balanced per-phase admittance and f_a, f_b, f_c the phase factors: a grounded star gives $Y_{\text{ph}} = \text{diag}(f_a y, f_b y, f_c y)$, or $(Z_{\text{diag}} + Z_n\mathbf{J})^{-1}$ with a neutral impedance, where $Z_{\text{diag}} = \text{diag}(1/f_p y)$ and $\mathbf{J}$ is the matrix of ones. An ungrounded star gives $\text{diag}(y_p) - y_p y_p^T / \sum y_p$. A delta of branches $f_{ab}y/3$, $f_{bc}y/3$, $f_{ca}y/3$ gives the corresponding nodal matrix.

A.6 Newton–Raphson

With $Y = G + jB$, $\theta_{kj} = \theta_k - \theta_j$ and $S_k = P_k + jQ_k = U_k\left(\sum_j Y_{kj}U_j\right)^*$:

$$P_k = U_k \sum_j U_j \left(G_{kj}\cos\theta_{kj} + B_{kj}\sin\theta_{kj}\right), \qquad Q_k = U_k \sum_j U_j \left(G_{kj}\sin\theta_{kj} - B_{kj}\cos\theta_{kj}\right).$$

The Jacobian blocks, for $k \neq j$ and on the diagonal:

$$\begin{aligned} H_{kj} &= U_k U_j \left(G_{kj}\sin\theta_{kj} - B_{kj}\cos\theta_{kj}\right), & H_{kk} &= -Q_k - B_{kk}U_k^2, \\ N_{kj} &= U_k U_j \left(G_{kj}\cos\theta_{kj} + B_{kj}\sin\theta_{kj}\right), & N_{kk} &= P_k + G_{kk}U_k^2, \\ J_{kj} &= -N_{kj}, & J_{kk} &= P_k - G_{kk}U_k^2, \\ L_{kj} &= H_{kj}, & L_{kk} &= Q_k - B_{kk}U_k^2. \end{aligned}$$

PowerLab evaluates them all at once from the complex derivatives of $S = U(YU)^*$, and the tests check them against finite differences and against these formulas. Rows of H and N belong to the PU and PQ buses, rows of J and L to the PQ buses; columns of H and J to the angles of the PU and PQ buses, columns of N and L to the magnitudes of the PQ buses.

Flows and losses. The power into branch ij at end i is $S_{ij} = U_i\,I_{ij}^*$ with I_{ij} from the branch's 2×2 block. The branch losses are $S_{ij} + S_{ji}$, and P_{loss} is the sum of their real parts. The balance $P_{\text{gen}} - P_{\text{load}} - P_{\text{loss}}$ closes at the solution.

A.7 Two-port matrices

$$F_{\text{series}}(z) = \begin{bmatrix} 1 & z \\ 0 & 1 \end{bmatrix}, \quad F_{\text{shunt}}(y) = \begin{bmatrix} 1 & 0 \\ y & 1 \end{bmatrix}, \quad F_{\pi} = F_{\text{shunt}}(y_{\text{sh}})\, F_{\text{series}}(z)\, F_{\text{shunt}}(y_{\text{sh}}),$$

cascaded in order along the path from port 1 to port 2. Parallel paths are merged into one equivalent section. From the Y-bus reduced onto the two ports, $\left[\begin{smallmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{smallmatrix}\right]$,

$$A = -\frac{Y_{22}}{Y_{21}}, \quad B = -\frac{1}{Y_{21}}, \quad C = -\frac{\det Y}{Y_{21}}, \quad D = -\frac{Y_{11}}{Y_{21}},$$

with I_2 counted as leaving port 2. The equivalent impedance with a termination Z_{term} at port 2 is $(AZ_{\text{term}} + B)/(CZ_{\text{term}} + D)$.

Appendix B

Keyboard and mouse

B.1 Keyboard shortcuts

Keys	Action
PROJECT	
Ctrl + N	New project
Ctrl + O	Open a project
Ctrl + S	Save
Ctrl + ⇧ + S	Save as
Ctrl + E	Export results
Ctrl + ⇧ + E	Export the diagram
Ctrl + Q	Exit
EDITING	
Ctrl + Z	Undo
Ctrl + Y or Ctrl + ⇧ + Z	Redo
Ctrl + X / Ctrl + C / Ctrl + V	Cut / copy / paste (canvas focused)
Ctrl + D	Duplicate the selection
Del	Delete the selection
Ctrl + A	Select all
Ctrl + F	Find an element by name or id
Ctrl + Alt + R	Rotate the selection 90°
Ctrl + Alt + O	Switch the selected buses between vertical and horizontal
Esc	Cancel a connection drag, or drop the component picked in the library
VIEW	
Ctrl + + / Ctrl + -	Zoom in / out
Ctrl + 0	Fit the diagram to the window
Ctrl + 1	Actual size
Ctrl + R	Show or hide the result annotations
Ctrl + ⇧ + D	Data manager
STUDIES	
F5	Run all study cases

B.2 Mouse

Gesture	Action
Click a library entry, then the canvas	Place a component (⇧ keeps the tool for several)
Drag a library entry onto the canvas	Place a component
Click an element	Select it
Ctrl + click	Add to or remove from the selection
Drag on empty canvas	Rubber-band selection
Drag an element	Move it (and the rest of the selection)
Double-click an element	Show its properties
Drag from a terminal	Connect it to a bus, or to a new bus on empty canvas
Drag a selected bus's end handle	Resize the bar
Drag a label	Move the label
Double-click a wire	Add a bend
Drag a selected wire's bend handle	Move the bend
Right-click an element or wire	Context menu
Right-click while placing or connecting	Cancel
Ctrl + wheel	Zoom around the cursor
Wheel	Scroll
Middle-button drag, or Space + drag	Pan

Appendix C

How PowerLab is verified

PowerLab computes engineering results, so it is tested against independent references rather than against its own earlier output. An automated test suite, a few hundred tests, runs in seconds and covers the engine, the file format, the command line and the window, which is driven without a screen.

C.1 References

MATLAB reference solutions. Every numeric variable of the MATLAB solutions of the four assignments, and of the LF variant, is stored at full precision (17 significant digits) for $A = 80$ and $A = 0$. The tests run every case of the example projects at both values of `A` and compare with a relative tolerance of 10^{-9}, and 10^{-8} for the load flow. A refactoring that moves a result in its eighth digit fails.

The assignment statements. The $A = 0$ check values printed in the statements are an independent check, not a comparison with anything PowerLab produced, for example $Z_I = 22.5156 + j11.0767$ pu in S1 and $P_{\text{loss}} = 26.4240$ MW in LF.

pandapower. The load flow is cross-checked against pandapower, an independent open-source solver, on the LF network: voltages, angles, slack power and losses agree to 10^{-8}. This test is opt-in (`PSA_ORACLES=1`), because importing pandapower takes longer than the rest of the suite.

Finite differences. The load-flow Jacobian is compared with a finite-difference Jacobian at a random state, and its diagonal with the textbook formulas of Section A.6.

C.2 Self-checks in every result

The results carry their own cross-checks (Chapter 9): the two-port matrix by cascade and by Kron reduction, the equivalent impedance by Kron reduction and through the two-port, the Thevenin impedance by the impedance matrix and by Kron reduction, the Thevenin voltage by the nodal solve and by the MATLAB route, $\det F = 1$ for reciprocal networks, the symmetry of the admittance matrix, and the load-flow power balance. A residual well above rounding level is a sign that something in the network is not what you think.

C.3 Properties that must hold

Besides fixed numbers, the tests check relations any correct solver satisfies. A few examples:

- an unbalanced load with phase factors $(1, 1, 1)$ gives the balanced solution;
- the general fault solver reproduces the closed-form currents of the three-phase, line-to-ground, line-to-line and double line-to-ground faults;
- a `Dyn` transformer is a shunt on its star side in the zero sequence, and nothing on its delta side;
- two generators sharing a bus give the same flow as one generator with their sum;
- a constant-impedance load draws $S_n U^2$ at the solved voltage.

C.4 Running the tests yourself

From the repository's root:

```
python -m pytest                            # the whole suite
python -m pytest tests/test_lf.py           # one file
PSA_ORACLES=1 python -m pytest tests/test_lf.py -k pandapower
python -m unittest discover -s tests -t .   # without pytest
```

The reference tables of Chapters 5 and 9 are generated from the program itself, by `tools/make_manual.py`. A test fails when they fall out of step with the code.